# CHAPTER 4

# *Multipoles, Electrostatics of Macroscopic Media, Dielectrics*

This chapter is first concerned with the potential due to localized charge distributions and its expansion in multipoles. The development is made in terms of spherical harmonics, but contact is established with the rectangular components for the first few multipoles. The energy of a multipole in an external field is then discussed. An elementary derivation of the macroscopic equations of electrostatics is sketched, but a careful treatment is deferred to Chapter 6. Dielectrics and the appropriate boundary conditions are then described, and some typical boundary-value problems with dielectrics are solved. Simple classical models are used to illustrate the main features of atomic polarizability and susceptibility. Finally the question of electrostatic energy and forces in the presence of dielectrics is discussed.

## *4.1 Multipole Expansion*

A localized distribution of charge is described by the charge density $\rho(\mathbf{x}')$, which is nonvanishing only inside a sphere of radius $R$ around some origin.* The potential outside the sphere can be written as an expansion in spherical harmonics:

$$\Phi(\mathbf{x}) = \frac{1}{4\pi\epsilon_0} \sum_{l=0}^{\infty} \sum_{m=-l}^{l} \frac{4\pi}{2l+1} q_{lm} \frac{Y_{lm}(\theta, \phi)}{r^{l+1}} \tag{4.1}$$

where the particular choice of constant coefficients is made for later convenience. Equation (4.1) is called a multipole expansion; the $l = 0$ term is called the monopole term, $l = 1$ are the dipole terms, etc. The reason for these names becomes clear below. The problem to be solved is the determination of the constants $q_{lm}$ in terms of the properties of the charge density $\rho(\mathbf{x}')$. The solution is very easily obtained from the integral (1.17) for the potential:

$$\Phi(\mathbf{x}) = \frac{1}{4\pi\epsilon_0} \int \frac{\rho(\mathbf{x}')}{|\mathbf{x} - \mathbf{x}'|} d^3x'$$

with expansion (3.70) for $1/|\mathbf{x} - \mathbf{x}'|$. Since we are interested at the moment in the potential outside the charge distribution, $r_< = r'$ and $r_> = r$. Then we find:

$$\Phi(\mathbf{x}) = \frac{1}{\epsilon_0} \sum_{l,m} \frac{1}{2l+1} \left[ \int Y^*_{lm}(\theta', \phi') r'^l \rho(\mathbf{x}') \, d^3x' \right] \frac{Y_{lm}(\theta, \phi)}{r^{l+1}} \tag{4.2}$$

*The sphere of radius $R$ is an arbitrary conceptual device employed merely to divide space into regions with and without charge. If the charge density falls off with distance faster than any power, the expansion in multipoles is valid at large enough distances.

Consequently the coefficients in (4.1) are:

$$q_{lm} = \int Y^*_{lm}(\theta', \phi') r'^l \rho(\mathbf{x}')\, d^3x' \tag{4.3}$$

These coefficients are called *multipole moments*. To see the physical interpretation of them we exhibit the first few explicitly in terms of Cartesian coordinates:

$$q_{00} = \frac{1}{\sqrt{4\pi}} \int \rho(\mathbf{x}')\, d^3x' = \frac{1}{\sqrt{4\pi}}\, q \tag{4.4}$$

$$\left.\begin{aligned} q_{11} &= -\sqrt{\frac{3}{8\pi}} \int (x' - iy')\rho(\mathbf{x}')\, d^3x' = -\sqrt{\frac{3}{8\pi}}\,(p_x - ip_y) \\ q_{10} &= \sqrt{\frac{3}{4\pi}} \int z'\rho(\mathbf{x}')\, d^3x' = \sqrt{\frac{3}{4\pi}}\, p_z \end{aligned}\right\} \tag{4.5}$$

$$\left.\begin{aligned} q_{22} &= \frac{1}{4}\sqrt{\frac{15}{2\pi}} \int (x' - iy')^2\rho(\mathbf{x}')\, d^3x' = \frac{1}{12}\sqrt{\frac{15}{2\pi}}\,(Q_{11} - 2iQ_{12} - Q_{22}) \\ q_{21} &= -\sqrt{\frac{15}{8\pi}} \int z'(x' - iy')\rho(\mathbf{x}')\, d^3x' = -\frac{1}{3}\sqrt{\frac{15}{8\pi}}\,(Q_{13} - iQ_{23}) \\ q_{20} &= \frac{1}{2}\sqrt{\frac{5}{4\pi}} \int (3z'^2 - r'^2)\rho(\mathbf{x}')\, d^3x' = \frac{1}{2}\sqrt{\frac{5}{4\pi}}\, Q_{33} \end{aligned}\right\} \tag{4.6}$$

Only the moments with $m \geq 0$ have been given, since (3.54) shows that for a real charge density the moments with $m < 0$ are related through

$$q_{l,-m} = (-1)^m q^*_{lm} \tag{4.7}$$

In equations (4.4)–(4.6), $q$ is the total charge, or monopole moment, $\mathbf{p}$ is the electric dipole moment:

$$\mathbf{p} = \int \mathbf{x}'\rho(\mathbf{x}')\, d^3x' \tag{4.8}$$

and $Q_{ij}$ is the traceless quadrupole moment tensor:

$$Q_{ij} = \int (3x'_i x'_j - r'^2\delta_{ij})\rho(\mathbf{x}')\, d^3x' \tag{4.9}$$

We see that the $l$th multipole coefficients [$(2l + 1)$ in number] are linear combinations of the corresponding multipoles expressed in rectangular coordinates. The expansion of $\Phi(\mathbf{x})$ in rectangular coordinates

$$\Phi(\mathbf{x}) = \frac{1}{4\pi\epsilon_0}\left[\frac{q}{r} + \frac{\mathbf{p}\cdot\mathbf{x}}{r^3} + \frac{1}{2}\sum_{i,j} Q_{ij}\frac{x_i x_j}{r^5} + \cdots\right] \tag{4.10}$$

by direct Taylor series expansion of $1/|\mathbf{x} - \mathbf{x}'|$ will be left as an exercise for the reader. It becomes increasingly cumbersome to continue the expansion in (4.10) beyond the quadrupole terms.

The electric field components for a given multipole can be expressed most

easily in terms of spherical coordinates. The negative gradient of a term in (4.1) with definite $l$, $m$ has spherical components:

$$E_r = \frac{(l+1)}{(2l+1)\epsilon_0} q_{lm} \frac{Y_{lm}(\theta, \phi)}{r^{l+2}}$$
$$E_\theta = -\frac{1}{(2l+1)\epsilon_0} q_{lm} \frac{1}{r^{l+2}} \frac{\partial}{\partial\theta} Y_{lm}(\theta, \phi) \tag{4.11}$$
$$E_\phi = -\frac{1}{(2l+1)\epsilon_0} q_{lm} \frac{1}{r^{l+2}} \frac{im}{\sin\theta} Y_{lm}(\rho, \phi)$$

$\partial Y_{lm}/\partial\theta$ and $Y_{lm}/\sin\theta$ can be expressed as linear combinations of other $Y_{lm}$'s, but the expressions are not particularly illuminating and so will be omitted. The proper way to describe a vector multipole field is by *vector* spherical harmonics, discussed in Chapter 9.

For a dipole $\mathbf{p}$ along the $z$ axis, the fields in (4.11) reduce to the familiar form:

$$E_r = \frac{2p\cos\theta}{4\pi\epsilon_0 r^3}$$
$$E_\theta = \frac{p\sin\theta}{4\pi\epsilon_0 r^3} \tag{4.12}$$
$$E_\phi = 0$$

These dipole fields can be written in vector form by recombining (4.12) or by directly operating with the gradient on the dipole term in (4.10). The result for the field at a point $\mathbf{x}$ due to a dipole $\mathbf{p}$ at the point $\mathbf{x}_0$ is:

$$\mathbf{E}(\mathbf{x}) = \frac{3\mathbf{n}(\mathbf{p}\cdot\mathbf{n}) - \mathbf{p}}{4\pi\epsilon_0|\mathbf{x} - \mathbf{x}_0|^3} \tag{4.13}$$

where $\mathbf{n}$ is a unit vector directed from $\mathbf{x}_0$ to $\mathbf{x}$.

There are two important remarks to be made. The first concerns the relationship of the Cartesian multipole moments like (4.8) to the spherical multipole moments (4.3). The former are $(l+1)(l+2)/2$ in number and for $l > 1$ are more numerous than the $(2l+1)$ spherical components. There is no contradiction here. The root of the differences lies in the different rotational transformation properties of the two types of multipole moments; the Cartesian tensors are reducible, the spherical, irreducible—see Problem 4.3. Note that for $l = 2$ we have recognized the difference by defining a *traceless* Cartesian quadrupole moment (4.9).

The second remark is that in general the multipole moment coefficients in the expansion (4.1) depend on the choice of origin. As a blatant example, consider a *point charge* $e$ located at $\mathbf{x}_0 = (r_0, \theta_0, \phi_0)$. Its potential has a multipole expansion of the form (4.1) with multipole moments,

$$q_{lm} = e r_0^l Y^*_{lm}(\theta_0, \phi_0)$$

These are nonvanishing for all $l$, $m$ in general. Only the $l = 0$ multipole $q_{00} = e/\sqrt{4\pi}$ is independent of the location of the point charge. For two point charges $+e$ and $-e$ at $\mathbf{x}_0$ and $\mathbf{x}_1$, respectively, the multipole moments are

$$q_{lm} = e[r_0^l Y^*_{lm}(\theta_0, \phi_0) - r_1^l Y^*_{lm}(\theta_1, \phi_1)]$$

Now the $l = 0$ multipole moment of the system vanishes, and the $l = 1$ moments are

$$q_{10} = \sqrt{\frac{3}{4\pi}}\, e(z_0 - z_1)$$

$$q_{11} = -\sqrt{\frac{3}{8\pi}}\, e[(x_0 - x_1) - i(y_0 - y_1)]$$

These moments are independent of the location of the origin, depending only on the relative position of the two charges, but all higher moments depend on the location of the origin as well. These simple examples are special cases of general theorem (see Problem 4.4). The values of $q_{lm}$ for the *lowest nonvanishing multipole moment* of any charge distribution are *independent of the choice of origin* of the coordinates, but all higher multipole moments do in general depend on the location of the origin.

Before leaving the general formulation of multipoles, we consider a result that is useful in elucidating the basic difference between electric and magnetic dipoles (see Section 5.6) as well as in other contexts. Consider a localized charge distribution $\rho(\mathbf{x})$ that gives rise to an electric field $\mathbf{E}(\mathbf{x})$ throughout space. We wish to calculate the integral of $\mathbf{E}$ over the *volume* of a sphere of radius $R$. We begin by examining the problem in general, but then specialize to the two extremes shown in Fig. 4.1, one in which the sphere contains all of the charge and the other in which the charge lies external to the sphere. Choosing the origin of coordinates at the center of the sphere, we have the volume integral of the electric field,

$$\int_{r<R} \mathbf{E}(\mathbf{x})\, d^3x = -\int_{r<R} \nabla\Phi\, d^3x \tag{4.14}$$

This can be converted to an integral over the surface of the sphere:

$$\int_{r<R} \mathbf{E}(\mathbf{x})\, d^3x = -\int_{r=R} R^2\, d\Omega\, \Phi(\mathbf{x})\mathbf{n} \tag{4.15}$$

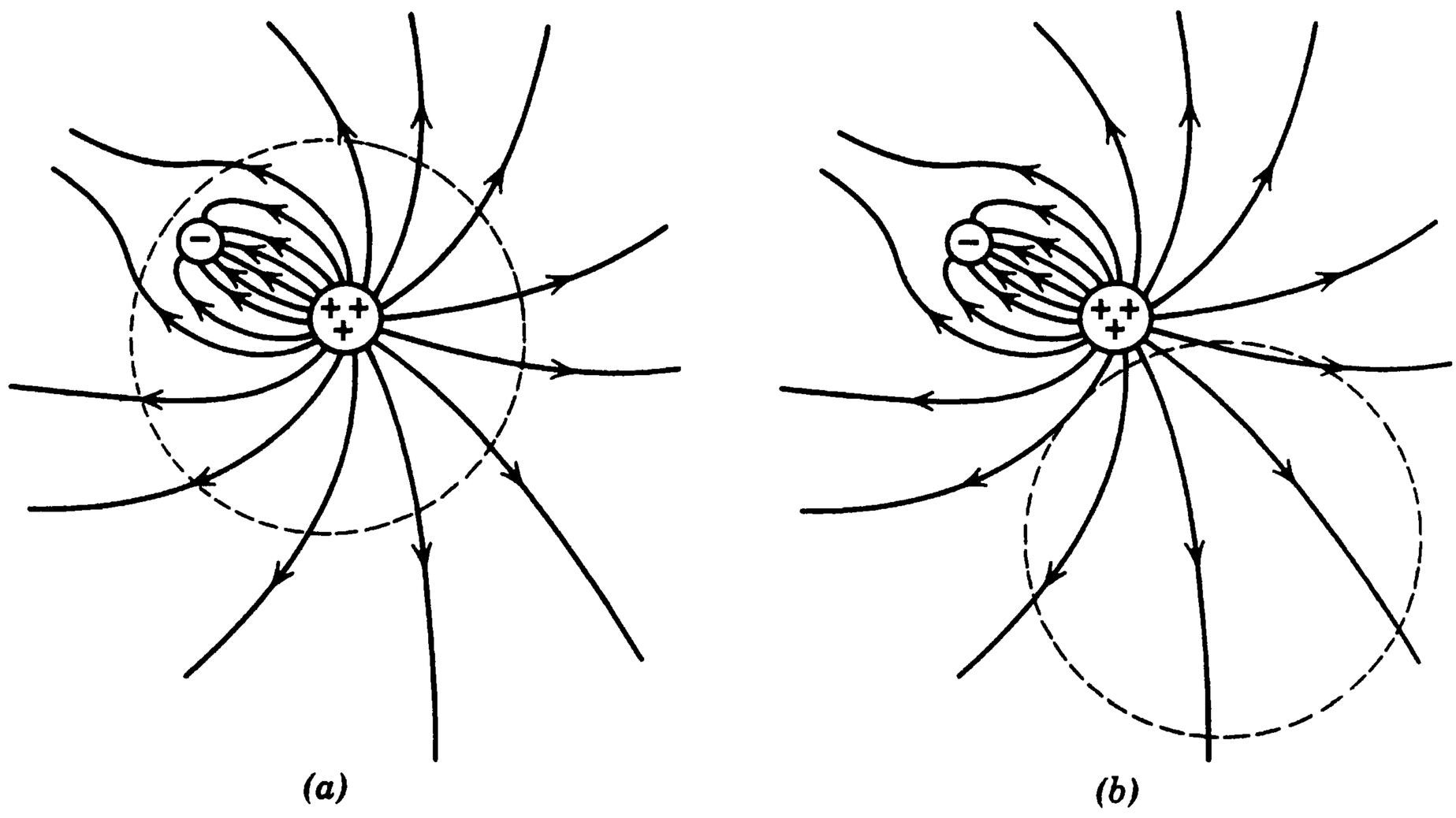



**Figure 4.1** Two configurations of charge density and the spheres within which the volume integral of electric field is to be calculated.

where $\mathbf{n}$ is the outwardly directed normal ($\mathbf{n} = \mathbf{x}/R$). Substitution of (1.17) for the potential leads to

$$\int_{r<R} \mathbf{E}(\mathbf{x})\, d^3x = -\frac{R^2}{4\pi\epsilon_0} \int d^3x'\, \rho(\mathbf{x}') \int_{r=R} d\Omega\, \frac{\mathbf{n}}{|\mathbf{x} - \mathbf{x}'|} \tag{4.16}$$

To perform the angular integration we first observe that $\mathbf{n}$ can be written in terms of the spherical angles $(\theta, \phi)$ as

$$\mathbf{n} = \mathbf{i} \sin\theta \cos\phi + \mathbf{j} \sin\theta \sin\phi + \mathbf{k} \cos\theta$$

Evidently the different components of $\mathbf{n}$ are linear combinations of $Y_{lm}$ for $l = 1$ *only*. When (3.38) or (3.70) is inserted into (4.16), orthogonality of the $Y_{lm}$ will eliminate all but the $l = 1$ term in the series. Thus we have

$$\int_{r=R} d\Omega\, \frac{\mathbf{n}}{|\mathbf{x} - \mathbf{x}'|} = \frac{r_<}{r_>^2} \int d\Omega\, \mathbf{n} \cos\gamma \tag{4.16$'$}$$

where $\cos\gamma = \cos\theta \cos\theta' + \sin\theta \sin\theta' \cos(\phi - \phi')$. The angular integral is equal to $4\pi\mathbf{n}'/3$, where $\mathbf{n}' = \mathbf{r}'/r'$. Thus the integral (4.16) is

$$\int_{r<R} \mathbf{E}(\mathbf{x})\, d^3x = -\frac{R^2}{3\epsilon_0} \int d^3x'\, \frac{r_<}{r_>^2}\, \mathbf{n}'\rho(\mathbf{x}') \tag{4.17}$$

where $(r_<, r_>) = (r', R)$ or $(R, r')$ depending on which of $r'$ and $R$ is larger.

If the sphere of radius $R$ completely encloses the charge density, as indicated in Fig. 4.1a, then $r_< = r'$ and $r_> = R$ in (4.17). The volume integral of the electric field over the sphere then becomes

$$\int_{r<R} \mathbf{E}(\mathbf{x})\, d^3x = -\frac{\mathbf{p}}{3\epsilon_0} \tag{4.18}$$

where $\mathbf{p}$ is the electric dipole moment (4.8) of the charge distribution with respect to the center of the sphere. Note that this volume integral is independent of the size of the *spherical* region of integration provided all the charge is inside.

If, on the other hand, the situation is as depicted in Fig. 4.1b, with the charge all exterior to the sphere of interest, $r_< = R$ and $r_> = r'$ in (4.17). Then we have

$$\int_{r<R} \mathbf{E}(\mathbf{x})\, d^3x = -\frac{R^3}{3\epsilon_0} \int d^3x'\, \frac{\mathbf{n}'}{r'^2}\, \rho(\mathbf{x}')$$

From Coulomb's law (1.5) the integral can be recognized to be the negative of $4\pi\epsilon_0$ times the electric field at the center of the sphere. Thus the volume integral of $\mathbf{E}$ is

$$\int_{r<R} \mathbf{E}(\mathbf{x})\, d^3x = \frac{4\pi}{3} R^3 \mathbf{E}(0) \tag{4.19}$$

In other words, the average value of the electric field over a spherical volume containing no charge is the value of the field at the center of the sphere.

The result (4.18) implies modification of (4.13) for the electric field of a dipole. To be consistent with (4.18), the dipole field must be written as

$$\mathbf{E}(\mathbf{x}) = \frac{1}{4\pi\epsilon_0} \left[ \frac{3\mathbf{n}(\mathbf{p} \cdot \mathbf{n}) - \mathbf{p}}{|\mathbf{x} - \mathbf{x}_0|^3} - \frac{4\pi}{3}\, \mathbf{p}\delta(\mathbf{x} - \mathbf{x}_0) \right] \tag{4.20}$$

The added delta function does not contribute to the field away from the site of the dipole. Its purpose is to yield the required volume integral (4.18), with the *convention* that the spherically symmetric (around $\mathbf{x}_0$) volume integral of the first term is zero (from angular integration), the singularity at $\mathbf{x} = \mathbf{x}_0$ causing an otherwise ambiguous result. Equation (4.20) and its magnetic dipole counterpart (5.64), when handled carefully, can be employed as if the dipoles were idealized point dipoles, the delta function terms carrying the essential information about the actually finite distributions of charge and current.

## 4.2 *Multipole Expansion of the Energy of a Charge Distribution in an External Field*

If a localized charge distribution described by $\rho(\mathbf{x})$ is placed in an *external* potential $\Phi(\mathbf{x})$, the electrostatic energy of the system is:

$$W = \int \rho(\mathbf{x})\Phi(\mathbf{x})\, d^3x \tag{4.21}$$

If the potential $\Phi$ is slowly varying over the region where $\rho(\mathbf{x})$ is nonnegligible, then it can be expanded in a Taylor series around a suitably chosen origin:

$$\Phi(\mathbf{x}) = \Phi(0) + \mathbf{x}\cdot\nabla\Phi(0) + \frac{1}{2}\sum_i\sum_j x_i x_j \frac{\partial^2\Phi}{\partial x_i\,\partial x_j}(0) + \cdots \tag{4.22}$$

Utilizing the definition of the electric field $\mathbf{E} = -\nabla\Phi$, the last two terms can be rewritten. Then (4.22) becomes:

$$\Phi(\mathbf{x}) = \Phi(0) - \mathbf{x}\cdot\mathbf{E}(0) - \frac{1}{2}\sum_i\sum_j x_i x_j \frac{\partial E_j}{\partial x_i}(0) + \cdots$$

Since $\nabla\cdot\mathbf{E} = 0$ for the external field, we can subtract

$$\tfrac{1}{6}r^2\nabla\cdot\mathbf{E}(0)$$

from the last term to obtain finally the expansion:

$$\Phi(\mathbf{x}) = \Phi(0) - \mathbf{x}\cdot\mathbf{E}(0) - \frac{1}{6}\sum_i\sum_j (3x_i x_j - r^2\delta_{ij})\frac{\partial E_j}{\partial x_i}(0) + \cdots \tag{4.23}$$

When this is inserted into (4.21) and the definitions of total charge, dipole moment (4.8), and quadrupole moment (4.9) are employed, the energy takes the form:

$$W = q\Phi(0) - \mathbf{p}\cdot\mathbf{E}(0) - \frac{1}{6}\sum_i\sum_j Q_{ij}\frac{\partial E_j}{\partial x_i}(0) + \cdots \tag{4.24}$$

This expansion shows the characteristic way in which the various multipoles interact with an external field—the charge with the potential, the dipole with the electric field, the quadrupole with the field gradient, and so on.

In nuclear physics the quadrupole interaction is of particular interest. Atomic nuclei can possess electric quadrupole moments, and their magnitudes and signs reflect the nature of the forces between neutrons and protons, as well as the

shapes of the nuclei themselves. The energy levels or states of a nucleus are described by the quantum numbers of total angular momentum $J$ and its projection $M$ along the $z$ axis, as well as others, which we will denote by a general index $\alpha$. A given nuclear state has associated with it a quantum-mechanical charge density* $\rho_{JM\alpha}(\mathbf{x})$, which depends on the quantum numbers $(J, M, \alpha)$ but is cylindrically symmetric about the $z$ axis. Thus the only nonvanishing quadrupole moment is $q_{20}$ in (4.6), or $Q_{33}$ in (4.9).† The quadrupole moment of a nuclear state is defined as the value of $(1/e)\, Q_{33}$ with the charge density $\rho_{JM\alpha}(\mathbf{x})$, where $e$ is the protonic charge:

$$Q_{JM\alpha} = \frac{1}{e}\int (3z^2 - r^2)\rho_{JM\alpha}(\mathbf{x})\, d^3x \tag{4.25}$$

The dimensions of $Q_{JM\alpha}$ are consequently (length)$^2$. Unless the circumstances are exceptional (e.g., nuclei in atoms with completely closed electronic shells), nuclei are subjected to electric fields that possess field gradients in the neighborhood of the nuclei. Consequently, according to (4.24), the energy of the nuclei will have a contribution from the quadrupole interaction. The states of different $M$ value for the same $J$ will have different quadrupole moments $Q_{JM\alpha}$, and so a degeneracy in $M$ value that may have existed will be removed by the quadrupole coupling to the "external" (crystal lattice, or molecular) electric field. Detection of these small energy differences by radiofrequency techniques allows the determination of the quadrupole moment of the nucleus.‡

The interaction energy between two dipoles $\mathbf{p}_1$ and $\mathbf{p}_2$ can be obtained directly from (4.24) by using the dipole field (4.20). Thus, the mutual potential energy is

$$W_{12} = \frac{\mathbf{p}_1 \cdot \mathbf{p}_2 - 3(\mathbf{n} \cdot \mathbf{p}_1)(\mathbf{n} \cdot \mathbf{p}_2)}{4\pi\epsilon_0 |\mathbf{x}_1 - \mathbf{x}_2|^3} \tag{4.26}$$

where $\mathbf{n}$ is a unit vector in the direction $(\mathbf{x}_1 - \mathbf{x}_2)$ and it is assumed that $\mathbf{x}_1 \neq \mathbf{x}_2$. The dipole-dipole interaction is attractive or repulsive, depending on the orientation of the dipoles. For fixed orientation and separation of the dipoles, the value of the interaction, averaged over the relative positions of the dipoles, is zero. If the moments are generally parallel, attraction (repulsion) occurs when the moments are oriented more or less parallel (perpendicular) to the line joining their centers. For antiparallel moments the reverse is true. The extreme values of the potential energy are equal in magnitude.

## 4.3 *Elementary Treatment of Electrostatics with Ponderable Media*

In Chapters 1, 2, and 3 we considered electrostatic potentials and fields in the presence of charges and conductors, but no other ponderable media. We there-

*See *Blatt and Weisskopf* (pp. 23 ff.) for an elementary discussion of the quantum aspects of the problem.

†Actually $Q_{11}$ and $Q_{22}$ are different from zero, but are not independent of $Q_{33}$, being given by $Q_{11} = Q_{22} = -\frac{1}{2}Q_{33}$.

‡"The quadrupole moment of a nucleus," denoted by $Q$, is defined as the value of $Q_{JM\alpha}$ in the state $M = J$. See *Blatt and Weisskopf, loc. cit.*

fore made no distinction between microscopic fields and macroscopic fields, although our treatment of conductors in an idealized fashion with *surface* charge densities implied a macroscopic description. Air is sufficiently tenuous that the neglect of its dielectric properties causes no great error; our results so far are applicable there. But much of electrostatics concerns itself with charges and fields in ponderable media whose respective electric responses must be taken into account. In the Introduction we indicated the need for averaging over macroscopically small, but microscopically large, regions to obtain the Maxwell equations appropriate for macroscopic phenomena. This is done in a careful fashion in Chapter 6, after the Maxwell equations with time variation have been discussed. For the present we merely remind the reader of the outlines of the elementary discussion of polarization in a fashion that glosses over difficult and sometimes subtle aspects of the averaging procedure and the introduction of the macroscopic quantities.

The first observation is that when an averaging is made of the homogeneous equation, $\nabla \times \mathbf{E}_{\text{micro}} = 0$, the same equation, namely,

$$\nabla \times \mathbf{E} = 0 \tag{4.27}$$

holds for the averaged, that is, the macroscopic, electric field $\mathbf{E}$. This means that the electric field is still derivable from a potential $\Phi(\mathbf{x})$ in electrostatics.

If an electric field is applied to a medium made up of a large number of atoms or molecules, the charges bound in each molecule will respond to the applied field and will execute perturbed motions. The molecular charge density will be distorted. The multipole moments of each molecule will be different from what they were in the absence of the field. In simple substances, when there is no applied field the multipole moments are all zero, at least when averaged over many molecules. The dominant molecular multipole with the applied fields is the dipole. There is thus produced in the medium an *electric polarization* $\mathbf{P}$ (dipole moment per unit volume) given by

$$\mathbf{P}(\mathbf{x}) = \sum_i N_i \langle \mathbf{p}_i \rangle \tag{4.28}$$

where $\mathbf{p}_i$ is the dipole moment of the $i$th type of molecule in the medium, the average is taken over a small volume centered at $\mathbf{x}$ and $N_i$ is the average number per unit volume of the $i$th type of molecule at the point $\mathbf{x}$. If the molecules have a net charge $e_i$ and, in addition, there is macroscopic excess or free charge, the charge density at the macroscopic level will be

$$\rho(\mathbf{x}) = \sum_i N_i \langle e_i \rangle + \rho_{\text{excess}} \tag{4.29}$$

Usually the average molecular charge is zero. Then the charge density is the excess or free charge (suitably averaged).

If we now look at the medium from a macroscopic point of view, we can build up the potential or field by linear superposition of the contributions from each macroscopically small volume element $\Delta V$ at the variable point $\mathbf{x}'$. Thus the charge of $\Delta V$ is $\rho(\mathbf{x}')\,\Delta V$ and the dipole moment of $\Delta V$ is $\mathbf{P}(\mathbf{x}')\,\Delta V$. If there are no higher macroscopic multipole moment densities, the potential $\Delta\Phi(\mathbf{x}, \mathbf{x}')$

caused by the configuration of moments in $\Delta V$ can be seen from (4.10) to be given without approximation by

$$\Delta\Phi(\mathbf{x}, \mathbf{x}') = \frac{1}{4\pi\epsilon_0}\left[\frac{\rho(\mathbf{x}')}{|\mathbf{x} - \mathbf{x}'|}\Delta V + \frac{\mathbf{P}(\mathbf{x}') \cdot (\mathbf{x} - \mathbf{x}')}{|\mathbf{x} - \mathbf{x}'|^3}\Delta V\right] \tag{4.30}$$

provided $\mathbf{x}$ is outside $\Delta V$. We now treat $\Delta V$ as (macroscopically) infinitesimal, put it equal to $d^3x'$, and integrate over all space to obtain the potential

$$\Phi(\mathbf{x}) = \frac{1}{4\pi\epsilon_0}\int d^3x'\left[\frac{\rho(\mathbf{x}')}{|\mathbf{x} - \mathbf{x}'|} + \mathbf{P}(\mathbf{x}') \cdot \nabla'\left(\frac{1}{|\mathbf{x} - \mathbf{x}'|}\right)\right] \tag{4.31}$$

The second term is analogous to the dipole layer potential (1.25), but is for a volume distribution of dipoles. An integration by parts transforms the potential into

$$\Phi(\mathbf{x}) = \frac{1}{4\pi\epsilon_0}\int d^3x' \frac{1}{|\mathbf{x} - \mathbf{x}'|}[\rho(\mathbf{x}') - \nabla' \cdot \mathbf{P}(\mathbf{x}')] \tag{4.32}$$

This is just the customary expression for the potential caused by a charge distribution $(\rho - \nabla \cdot \mathbf{P})$. With $\mathbf{E} = -\nabla\Phi$, the first Maxwell equation therefore reads

$$\nabla \cdot \mathbf{E} = \frac{1}{\epsilon_0}[\rho - \nabla \cdot \mathbf{P}] \tag{4.33}$$

The presence of the divergence of $\mathbf{P}$ in the effective charge density can be understood qualitatively. If the polarization is nonuniform there can be a net increase or decrease of charge within any small volume, as indicated schematically in Fig. 4.2.

With the definition of the *electric displacement* $\mathbf{D}$,

$$\mathbf{D} = \epsilon_0\mathbf{E} + \mathbf{P} \tag{4.34}$$

(4.33) becomes the familiar

$$\nabla \cdot \mathbf{D} = \rho \tag{4.35}$$

Equations (4.27) and (4.35) are the macroscopic counterparts of (1.13) and (1.14) of Chapter 1.

As discussed in the Introduction, a *constitutive relation* connecting $\mathbf{D}$ and $\mathbf{E}$ is necessary before a solution for the electrostatic potential or fields can be obtained. In the subsequent sections of this chapter we assume that the response of the system to an applied field is linear. This excludes ferroelectricity from discussion, but otherwise is no real restriction provided the field strengths do not

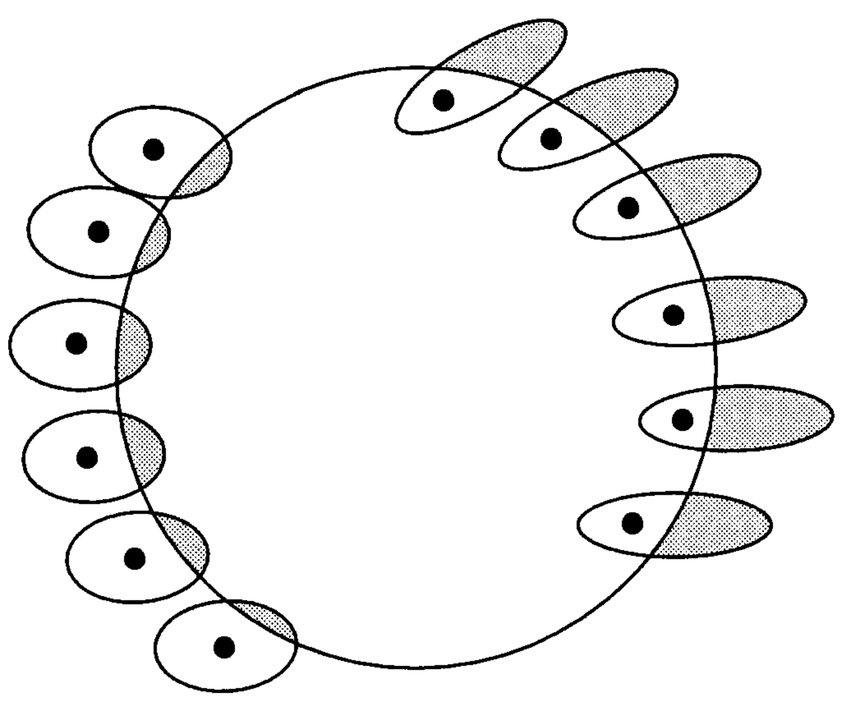


**Figure 4.2** Origin of polarization-charge density. Because of spatial variation of polarization, more molecular charge may leave a given small volume than enters it. Only molecules near the boundary are shown.

become extremely large. As a further simplification we suppose that the *medium is isotropic*. Then the induced polarization **P** is parallel to **E** with a coefficient of proportionality that is independent of direction:

$$\mathbf{P} = \epsilon_0 \chi_e \mathbf{E} \tag{4.36}$$

The constant $\chi_e$ is called the *electric susceptibility* of the medium. The displacement **D** is therefore proportional to **E**,

$$\mathbf{D} = \epsilon \mathbf{E} \tag{4.37}$$

where

$$\epsilon = \epsilon_0(1 + \chi_e) \tag{4.38}$$

is the electric permittivity; $\epsilon/\epsilon_0 = 1 + \chi_e$ is called the *dielectric constant* or relative electric permittivity.

If the dielectric is not only isotropic, but also uniform, then $\epsilon$ is independent of position. The divergence equations (4.35) can then be written

$$\nabla \cdot \mathbf{E} = \rho/\epsilon \tag{4.39}$$

All problems *in that medium* are reduced to those of preceding chapters, except that the electric fields produced by given charges are reduced by a factor $\epsilon_0/\epsilon$. The reduction can be understood in terms of a polarization of the atoms that produce fields in opposition to that of the given charge. One immediate consequence is that the capacitance of a capacitor is increased by a factor of $\epsilon/\epsilon_0$ if the empty space between the electrodes is filled with a dielectric with dielectric constant $\epsilon/\epsilon_0$ (true only to the extent that fringing fields can be neglected).

If the uniform medium does not fill all of the space where there are electric fields or, more generally, if there are different media juxtaposed, not necessarily linear in their responses, we must consider the question of boundary conditions on **D** and **E** at the interfaces between media. These boundary conditions are derived from the full set of Maxwell equations in Section I.5. The results are that the normal components of **D** and the tangential components of **E** on either side of an interface satisfy the *boundary conditions*, valid for time-varying as well as static fields,

$$\left.\begin{aligned} (\mathbf{D}_2 - \mathbf{D}_1) \cdot \mathbf{n}_{21} &= \sigma \\ (\mathbf{E}_2 - \mathbf{E}_1) \times \mathbf{n}_{21} &= 0 \end{aligned}\right\} \tag{4.40}$$

where $\mathbf{n}_{21}$ is a unit normal to the surface, directed from region 1 to region 2, and $\sigma$ is the macroscopic surface-charge density on the boundary surface (*not* including the polarization charge).

## *4.4 Boundary-Value Problems with Dielectrics*

The methods of earlier chapters for the solution of electrostatic boundary-value problems can readily be extended to handle the presence of dielectrics. In this section we treat a few examples of the various techniques applied to dielectric media.

To illustrate the method of images for dielectrics we consider a point charge $q$ embedded in a semi-infinite dielectric $\epsilon_1$ a distance $d$ away from a plane inter-

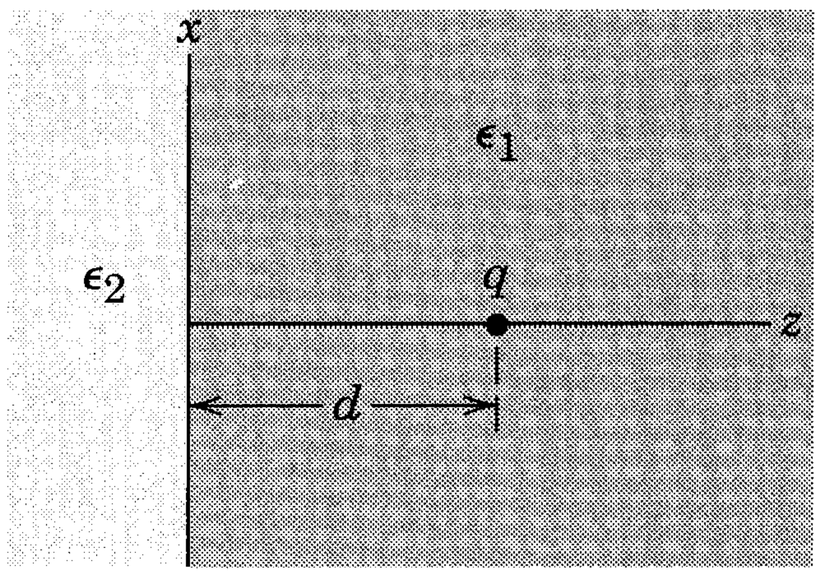



Figure 4.3

face that separates the first medium from another semi-infinite dielectric $\epsilon_2$. The surface may be taken as the plane $z = 0$, as shown in Fig. 4.3. We must find the appropriate solution to the equations:

$$\begin{aligned} \epsilon_1 \nabla \cdot \mathbf{E} &= \rho, \qquad z > 0 \\ \epsilon_2 \nabla \cdot \mathbf{E} &= 0, \qquad z < 0 \end{aligned} \tag{4.41}$$

and

$$\nabla \times \mathbf{E} = 0, \qquad \text{everywhere}$$

subject to the boundary conditions at $z = 0$:

$$\lim_{z \to 0^+} \begin{Bmatrix} \epsilon_1 E_z \\ E_x \\ E_y \end{Bmatrix} = \lim_{z \to 0^-} \begin{Bmatrix} \epsilon_2 E_z \\ E_x \\ E_y \end{Bmatrix} \tag{4.42}$$

Since $\nabla \times \mathbf{E} = 0$ everywhere, $\mathbf{E}$ is derivable in the usual way from a potential $\Phi$. In attempting to use the image method it is natural to locate an image charge $q'$ at the symmetrical position $A'$ shown in Fig. 4.4. Then for $z > 0$ the potential at a point $P$ described by cylindrical coordinates $(\rho, \phi, z)$ will be

$$\Phi = \frac{1}{4\pi\epsilon_1}\left(\frac{q}{R_1} + \frac{q'}{R_2}\right), \qquad z > 0 \tag{4.43}$$

where $R_1 = \sqrt{\rho^2 + (d - z)^2}$, $R_2 = \sqrt{\rho^2 + (d + z)^2}$. So far the procedure is completely analogous to the problem with a conducting material in place of the dielectric $\epsilon_2$ for $z < 0$. But we now must specify the potential for $z < 0$. Since there are no charges in the region $z < 0$, it must be a solution of the Laplace equation without singularities in that region. Clearly the simplest assumption is that for $z < 0$ the potential is equivalent to that of a charge $q''$ at the position $A$ of the actual charge $q$:

$$\Phi = \frac{1}{4\pi\epsilon_2}\frac{q''}{R_1}, \qquad z < 0 \tag{4.44}$$

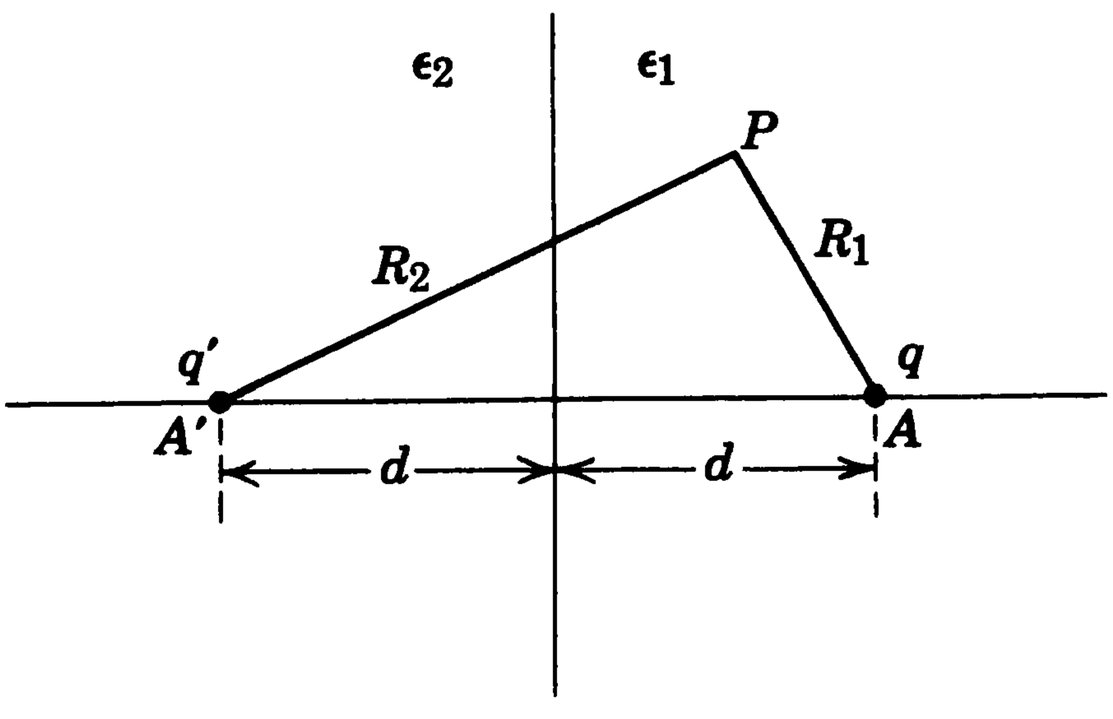



Figure 4.4

Since

$$\frac{\partial}{\partial z}\left(\frac{1}{R_1}\right)\bigg|_{z=0} = -\frac{\partial}{\partial z}\left(\frac{1}{R_2}\right)\bigg|_{z=0} = \frac{d}{(\rho^2 + d^2)^{3/2}}$$

while

$$\frac{\partial}{\partial \rho}\left(\frac{1}{R_1}\right)\bigg|_{z=0} = \frac{\partial}{\partial \rho}\left(\frac{1}{R_2}\right)\bigg|_{z=0} = \frac{-\rho}{(\rho^2 + d^2)^{3/2}}$$

the boundary conditions (4.42) lead to the requirements:

$$q - q' = q''$$
$$\frac{1}{\epsilon_1}(q + q') = \frac{1}{\epsilon_2} q''$$

These can be solved to yield the image charges $q'$ and $q''$:

$$q' = -\left(\frac{\epsilon_2 - \epsilon_1}{\epsilon_2 + \epsilon_1}\right)q$$
$$q'' = \left(\frac{2\epsilon_2}{\epsilon_2 + \epsilon_1}\right)q \tag{4.45}$$

For the two cases $\epsilon_2 > \epsilon_1$ and $\epsilon_2 < \epsilon_1$ the lines of force (actually lines of $\mathbf{D}$) are shown qualitatively in Fig. 4.5.

The polarization-charge density is given by $-\nabla \cdot \mathbf{P}$. Inside either dielectric, $\mathbf{P} = \epsilon_0 \chi_e \mathbf{E}$, so that $-\nabla \cdot \mathbf{P} = -\epsilon_0 \chi_e \nabla \cdot \mathbf{E} = 0$, except at the point charge $q$. At the surface, however, $\chi_e$ takes a discontinuous jump, $\nabla \chi_e = (\epsilon_1 - \epsilon_2)/\epsilon_0$ as $z$ passes through $z = 0$. This implies that there is a polarization-surface-charge density on the plane $z = 0$:

$$\sigma_{\text{pol}} = -(\mathbf{P}_2 - \mathbf{P}_1) \cdot \mathbf{n}_{21} \tag{4.46}$$

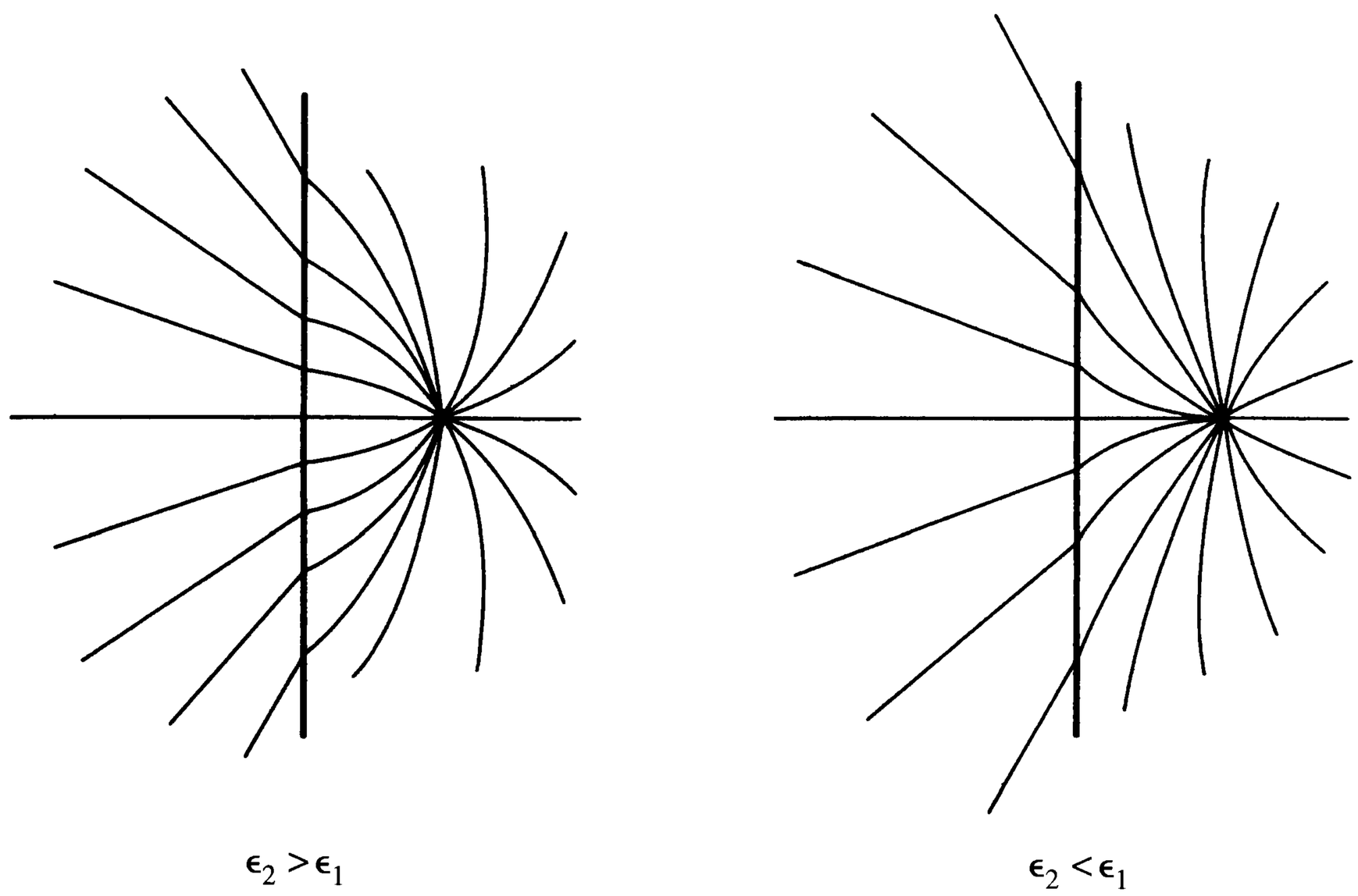



**Figure 4.5** Lines of electric displacement for a point charge embedded in a dielectric $\epsilon_1$ near a semi-infinite slab of dielectric $\epsilon_2$.

where $\mathbf{n}_{21}$ is the unit normal from dielectric 1 to dielectric 2, and $\mathbf{P}_i$ is the polarization in the dielectric $i$ at $z = 0$. Since

$$\mathbf{P}_i = (\epsilon_i - \epsilon_0)\mathbf{E}_i = -(\epsilon_i - \epsilon_0)\nabla\Phi(0^{\pm})$$

it is a simple matter to show that the polarization-charge density is

$$\sigma_{\text{pol}} = -\frac{q}{2\pi}\frac{\epsilon_0(\epsilon_2 - \epsilon_1)}{\epsilon_1(\epsilon_2 + \epsilon_1)}\frac{d}{(\rho^2 + d^2)^{3/2}} \tag{4.47}$$

In the limit $\epsilon_2 \gg \epsilon_1$ the dielectric $\epsilon_2$ behaves much like a conductor in that the electric field inside it becomes very small and the surface-charge density (4.47) approaches the value appropriate to a conducting surface, apart from a factor of $\epsilon_0/\epsilon_1$.

The second illustration of electrostatic problems involving dielectrics is that of a dielectric sphere of radius $a$ with dielectric constant $\epsilon/\epsilon_0$ placed in an initially uniform electric field, which at large distances from the sphere is directed along the $z$ axis and has magnitude $E_0$, as indicated in Fig. 4.6. Both inside and outside the sphere there are no free charges. Consequently the problem is one of solving the Laplace equation with the proper boundary conditions at $r = a$. From the axial symmetry of the geometry we can take the solution to be of the form:

INSIDE:

$$\Phi_{\text{in}} = \sum_{l=0}^{\infty} A_l r^l P_l(\cos\theta) \tag{4.48}$$

OUTSIDE:

$$\Phi_{\text{out}} = \sum_{l=0}^{\infty} [B_l r^l + C_l r^{-(l+1)}]P_l(\cos\theta) \tag{4.49}$$

From the boundary condition at infinity ($\Phi \to -E_0 z = -E_0 r\cos\theta$) we find that the only nonvanishing $B_l$ is $B_1 = -E_0$. The other coefficients are determined from the boundary conditions at $r = a$:

TANGENTIAL $E$:

$$-\frac{1}{a}\frac{\partial\Phi_{\text{in}}}{\partial\theta}\bigg|_{r=a} = -\frac{1}{a}\frac{\partial\Phi_{\text{out}}}{\partial\theta}\bigg|_{r=a}$$

NORMAL $D$:

$$-\epsilon\frac{\partial\Phi_{\text{in}}}{\partial r}\bigg|_{r=a} = -\epsilon_0\frac{\partial\Phi_{\text{out}}}{\partial r}\bigg|_{r=a} \tag{4.50}$$

When the series (4.48) and (4.49) are substituted, there result two series of Legendre functions equal to zero. Since these must vanish for all $\theta$, the coef-

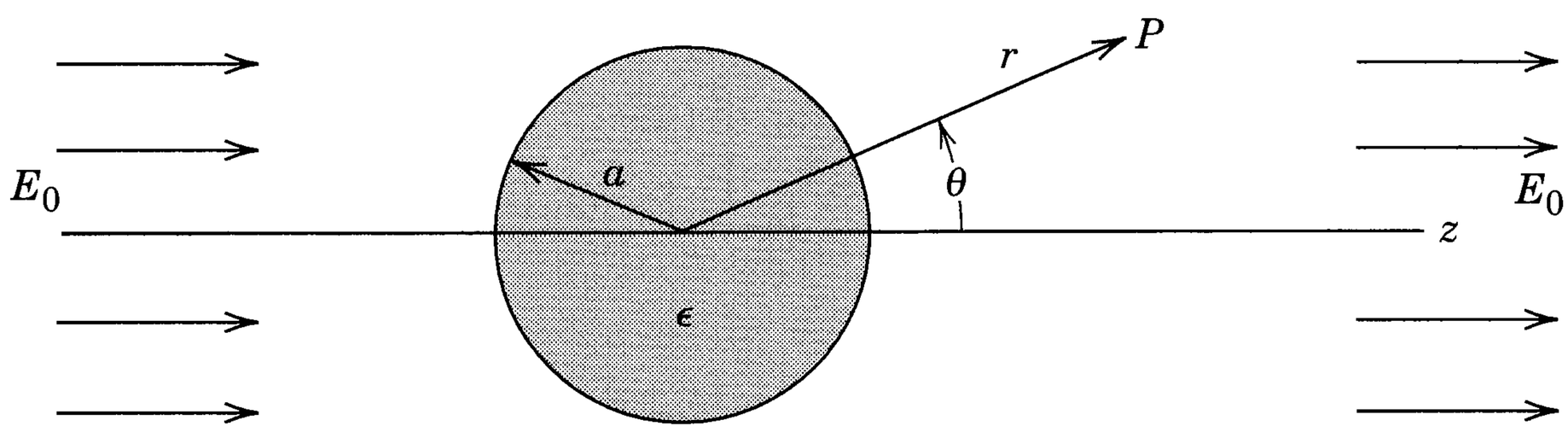



**Figure 4.6**

ficient of each Legendre function must vanish separately. For the first boundary condition this leads (through orthogonality of $P^1{}_l = \partial P_l/\partial\theta$) to the relations:

$$A_1 = -E_0 + \frac{C_1}{a^3}$$
$$A_l = \frac{C_l}{a^{2l+1}} \qquad \text{for } l \neq 1 \tag{4.51}$$

while the second gives (through orthogonality of $P_l$):

$$(\epsilon/\epsilon_0)A_1 = -E_0 - 2\frac{C_1}{a^3}$$
$$(\epsilon/\epsilon_0)lA_l = -(l+1)\frac{C_l}{a^{2l+1}} \qquad \text{for } l \neq 1 \tag{4.52}$$

The second equations in (4.51) and (4.52) can be satisfied simultaneously only with $A_l = C_l = 0$ for all $l \neq 1$. The remaining coefficients are given in terms of the applied electric field $E_0$:

$$A_1 = -\left(\frac{3}{2 + \epsilon/\epsilon_0}\right)E_0$$
$$C_1 = \left(\frac{\epsilon/\epsilon_0 - 1}{\epsilon/\epsilon_0 + 2}\right)a^3E_0 \tag{4.53}$$

The potential is therefore

$$\Phi_{\text{in}} = -\left(\frac{3}{\epsilon/\epsilon_0 + 2}\right)E_0 r\cos\theta$$
$$\Phi_{\text{out}} = -E_0 r\cos\theta + \left(\frac{\epsilon/\epsilon_0 - 1}{\epsilon/\epsilon_0 + 2}\right)E_0\frac{a^3}{r^2}\cos\theta \tag{4.54}$$

The potential inside the sphere describes a constant electric field parallel to the applied field with magnitude

$$E_{\text{in}} = \frac{3}{\epsilon/\epsilon_0 + 2}E_0 < E_0 \text{ if } \epsilon > \epsilon_0 \tag{4.55}$$

Outside the sphere the potential is equivalent to the applied field $E_0$ plus the field of an electric dipole at the origin with dipole moment:

$$p = 4\pi\epsilon_0\left(\frac{\epsilon/\epsilon_0 - 1}{\epsilon/\epsilon_0 + 2}\right)a^3E_0 \tag{4.56}$$

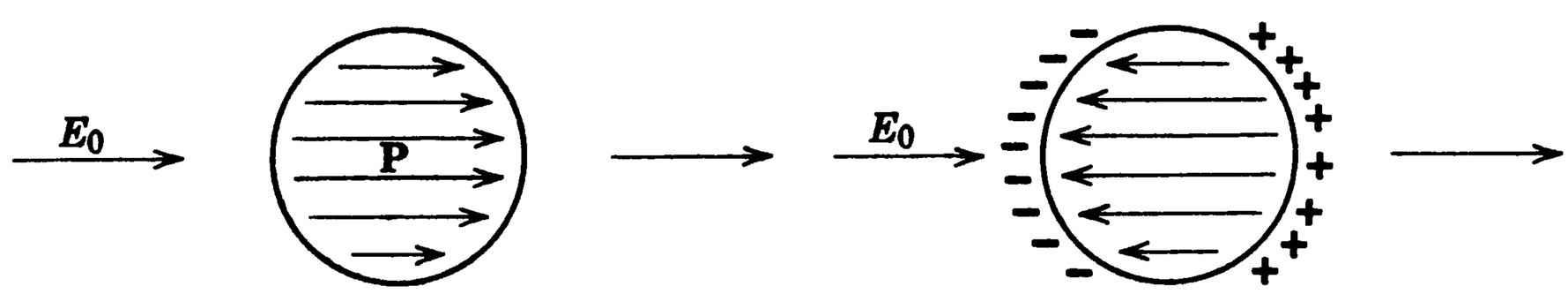



**Figure 4.7** Dielectric sphere in a uniform field $E_0$, showing the polarization on the left and the polarization charge with its associated, opposing, electric field on the right.

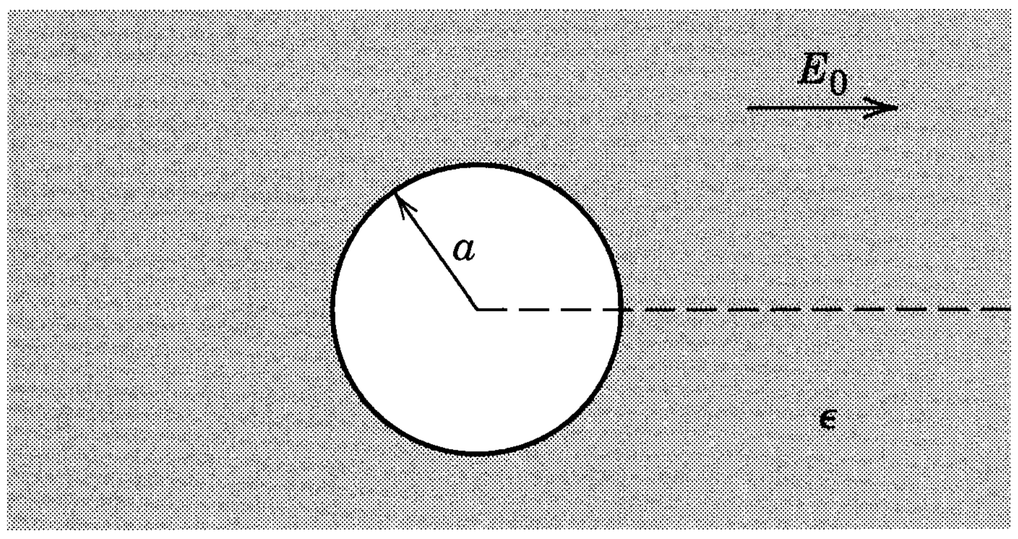



**Figure 4.8** Spherical cavity in a dielectric with a uniform field applied.

oriented in the direction of the applied field. The dipole moment can be interpreted as the volume integral of the polarization **P**. The polarization is

$$\mathbf{P} = (\epsilon - \epsilon_0)\mathbf{E} = 3\epsilon_0\left(\frac{\epsilon/\epsilon_0 - 1}{\epsilon/\epsilon_0 + 2}\right)\mathbf{E}_0 \tag{4.57}$$

It is constant throughout the volume of the sphere and has a volume integral given by (4.56). The polarization-surface-charge density is, according to (4.46), $\sigma_{\text{pol}} = (\mathbf{P}\cdot\mathbf{r})/r$:

$$\sigma_{\text{pol}} = 3\epsilon_0\left(\frac{\epsilon/\epsilon_0 - 1}{\epsilon/\epsilon_0 + 2}\right)E_0 \cos\theta \tag{4.58}$$

This can be thought of as producing an internal field directed oppositely to the applied field, so reducing the field inside the sphere to its value (4.55), as sketched in Fig. 4.7.

The problem of a spherical cavity of radius $a$ in a dielectric medium with dielectric constant $\epsilon/\epsilon_0$ and with an applied electric field $E_0$ parallel to the $z$ axis, as shown in Fig. 4.8, can be handled in exactly the same way as the dielectric sphere. In fact, inspection of boundary conditions (4.50) shows that the results for the cavity can be obtained from those of the sphere by the replacement $\epsilon/\epsilon_0 \to (\epsilon_0/\epsilon)$. Thus, for example, the field inside the cavity is uniform, parallel to $\mathbf{E}_0$, and of magnitude:

$$E_{\text{in}} = \frac{3\epsilon}{2\epsilon + \epsilon_0}E_0 > E_0 \text{ if } \epsilon > \epsilon_0 \tag{4.59}$$

Similarly, the field outside is the applied field plus that of a dipole at the origin *oriented oppositely* to the applied field and with dipole moment:

$$p = 4\pi\epsilon_0\left(\frac{\epsilon/\epsilon_0 - 1}{2\epsilon/\epsilon_0 + 1}\right)a^3E_0 \tag{4.60}$$

## *4.5 Molecular Polarizability and Electric Susceptibility*

In this section and the next we consider the relation between molecular properties and the macroscopically defined parameter, the electric susceptibility $\chi_e$. Our discussion is in terms of simple classical models of the molecular properties, although a proper treatment necessarily would involve quantum-mechanical considerations. Fortunately, the simpler properties of dielectrics are amenable to classical analysis.

Before examining how the detailed properties of the molecules are related to the susceptibility, we must make a distinction between the fields acting on the molecules in the medium and the applied field. The susceptibility is defined through the relation $\mathbf{P} = \epsilon_0 \chi_e \mathbf{E}$, where $\mathbf{E}$ is the macroscopic electric field. In rarefied media where molecular separations are large there is little difference between the macroscopic field and that acting on any molecule or group of molecules. But in dense media with closely packed molecules the polarization of neighboring molecules gives rise to an internal field $\mathbf{E}_i$ at any given molecule in addition to the average macroscopic field $\mathbf{E}$, so that the total field at the molecule is $\mathbf{E} + \mathbf{E}_i$. The internal field $\mathbf{E}_i$ can be written as the difference of two terms,

$$\mathbf{E}_i = \mathbf{E}_{\text{near}} - \mathbf{E}_P \tag{4.61}$$

where $\mathbf{E}_{\text{near}}$ is the actual contribution of the molecules close to the given molecule and $\mathbf{E}_P$ is the contribution from those molecules treated in an average continuum approximation described by the polarization $\mathbf{P}$. What we are saying here is that close to the molecule in question we must take care to recognize the specific atomic configuration and locations of the nearby molecules. Inside some macroscopically small, but microscopically large, volume $V$ we therefore subtract out the smoothed macroscopic equivalent of the nearby molecular contributions ($\mathbf{E}_P$) and replace it with the correctly evaluated contribution ($\mathbf{E}_{\text{near}}$). This difference is the extra internal field $\mathbf{E}_i$.

The result (4.18) for the integral of the electric field inside a spherical volume of radius $R$ containing a charge distribution can be used to calculate $\mathbf{E}_P$. If the volume $V$ is chosen to be a sphere of radius $R$ containing many molecules, the total dipole moment inside is

$$\mathbf{p} = \frac{4\pi R^3}{3} \mathbf{P}$$

provided $V$ is so small that $\mathbf{P}$ is essentially constant throughout the volume. Then (4.18) shows that the average electric field inside the sphere (just what is desired for $\mathbf{E}_P$) is

$$\mathbf{E}_P = \frac{3}{4\pi R^3} \int_{r<R} \mathbf{E}\, d^3\mathbf{x} = -\frac{\mathbf{P}}{3\epsilon_0} \tag{4.62}$$

The internal field can therefore be written

$$\mathbf{E}_i = \frac{1}{3\epsilon_0} \mathbf{P} + \mathbf{E}_{\text{near}} \tag{4.63}$$

The field due to the molecules near by is more difficult to determine. *Lorentz* (p. 138) showed that for atoms in a simple cubic lattice $\mathbf{E}_{\text{near}}$ vanishes at any lattice site. The argument depends on the symmetry of the problem, as can be seen as follows. Suppose that inside the sphere we have a cubic array of dipoles such as are shown in Fig. 4.9, with all their moments constant in magnitude and oriented along the same direction (remember that the sphere is macroscopically small even though it contains very many molecules). The positions of the dipoles are given by the coordinates $\mathbf{x}_{ijk}$ with the components along the coordinate axes ($ia$, $ja$, $ka$), where $a$ is the lattice spacing, and $i$, $j$, $k$ each take on positive and

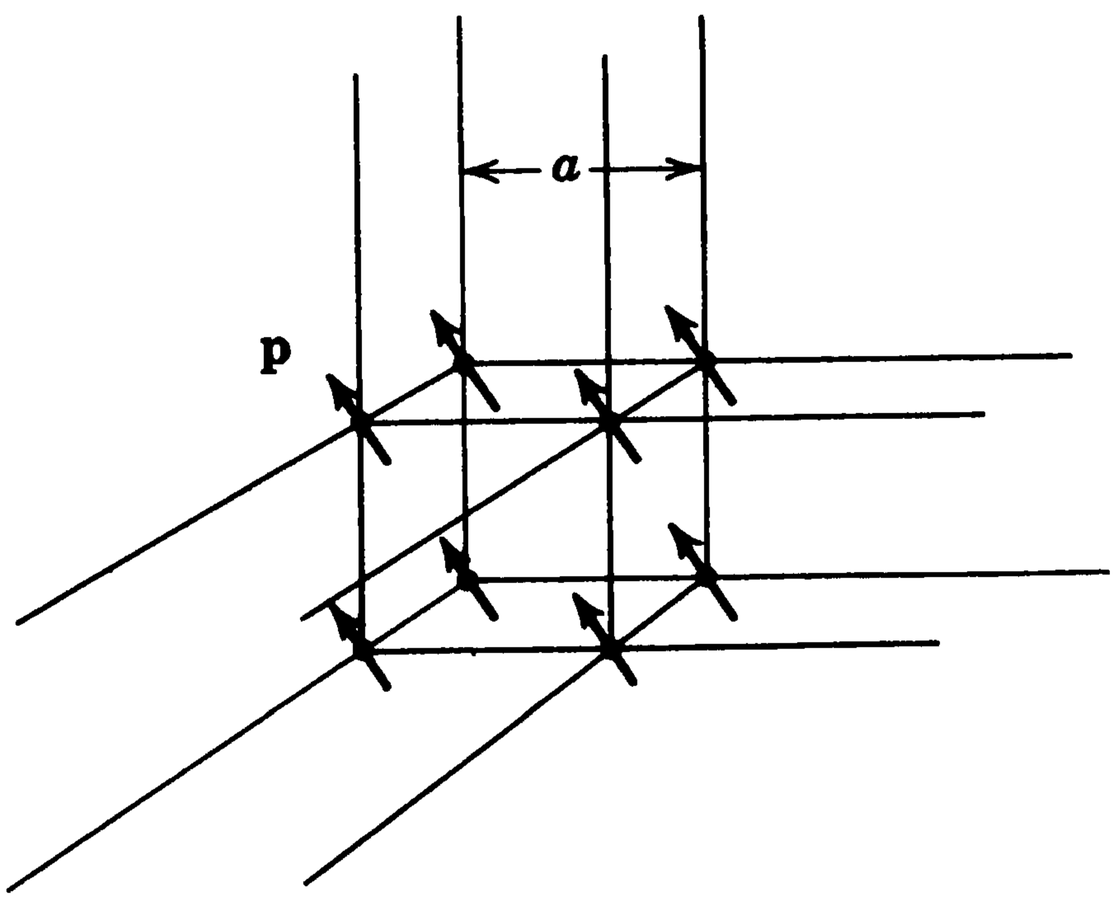



**Figure 4.9** Calculation of the internal field: contribution from nearby molecules in a simple cubic lattice.

negative integer values. The field at the origin due to all the dipoles is, according to (4.13),

$$\mathbf{E} = \sum_{i,j,k} \frac{3(\mathbf{p} \cdot \mathbf{x}_{ijk})\mathbf{x}_{ijk} - x_{ijk}^2 \mathbf{p}}{4\pi\epsilon_0 x_{ijk}^5} \tag{4.64}$$

The $x$ component of the field can be written in the form:

$$E_1 = \sum_{ijk} \frac{3(i^2 p_1 + ijp_2 + ikp_3) - (i^2 + j^2 + k^2)p_1}{4\pi\epsilon_0 a^3 (i^2 + j^2 + k^2)^{5/2}} \tag{4.65}$$

Since the indices run equally over positive and negative values, the cross terms involving $(ijp_2 + ikp_3)$ vanish. By symmetry the sums

$$\sum_{ijk} \frac{i^2}{(i^2 + j^2 + k^2)^{5/2}} = \sum_{ijk} \frac{j^2}{(i^2 + j^2 + k^2)^{5/2}} = \sum_{ijk} \frac{k^2}{(i^2 + j^2 + k^2)^{5/2}}$$

are all equal. Consequently

$$E_1 = \sum_{ijk} \frac{[3i^2 - (i^2 + j^2 + k^2)]p_1}{4\pi\epsilon_0 a^3 (i^2 + j^2 + k^2)^{5/2}} = 0 \tag{4.66}$$

Similar arguments show that the $y$ and $z$ components vanish also. Hence $\mathbf{E}_{\text{near}} = 0$ for a simple cubic lattice.

If $\mathbf{E}_{\text{near}} = 0$ for a highly symmetric situation, it seems plausible that $\mathbf{E}_{\text{near}} = 0$ also for completely random situations. Hence we expect amorphous substances to have no internal field due to nearby molecules. For lattices other than simple cubic, the components of $\mathbf{E}_{\text{near}}$ are related to the components of $\mathbf{P}$ through a traceless tensor $s_{\alpha\beta}$ that has the symmetry properties of the lattice. Nevertheless, it is a good working assumption that $\mathbf{E}_{\text{near}} \simeq 0$ for most materials.

The polarization vector $\mathbf{P}$ was defined in (4.28) as

$$\mathbf{P} = N\langle \mathbf{p}_{\text{mol}} \rangle$$

where $\langle \mathbf{p}_{\text{mol}} \rangle$ is the average dipole moment of the molecules. This dipole moment is approximately proportional to the electric field acting on the molecule. To exhibit this dependence on electric field we define the *molecular polarizability* $\gamma_{\text{mol}}$ as the ratio of the average molecular dipole moment to $\epsilon_0$ times the applied field at the molecule. Taking account of the internal field (4.63), this gives:

$$\langle \mathbf{p}_{\text{mol}} \rangle = \epsilon_0 \gamma_{\text{mol}} (\mathbf{E} + \mathbf{E}_i) \tag{4.67}$$

$\gamma_{\text{mol}}$ is, in principle, a function of the electric field, but for a wide range of field strengths is a constant that characterizes the response of the molecules to an applied field. Equation (4.67) can be combined with (4.28) and (4.63) to yield:

$$\mathbf{P} = N\gamma_{\text{mol}}\left(\epsilon_0\mathbf{E} + \frac{1}{3}\mathbf{P}\right) \tag{4.68}$$

where we have assumed $\mathbf{E}_{\text{near}} = 0$. Solving for $\mathbf{P}$ in terms of $\mathbf{E}$ and using the fact that $\mathbf{P} = \epsilon_0\chi_e\mathbf{E}$ defines the electric susceptibility of a substance, we find

$$\chi_e = \frac{N\gamma_{\text{mol}}}{1 - \frac{1}{3}N\gamma_{\text{mol}}} \tag{4.69}$$

as the relation between susceptibility (the macroscopic parameter) and molecular polarizability (the microscopic parameter). Since the dielectric constant is $\epsilon/\epsilon_0 = 1 + \chi_e$, it can be expressed in terms of $\gamma_{\text{mol}}$, or alternatively the molecular polarizability can be expressed in terms of the dielectric constant:

$$\gamma_{\text{mol}} = \frac{3}{N}\left(\frac{\epsilon/\epsilon_0 - 1}{\epsilon/\epsilon_0 + 2}\right) \tag{4.70}$$

This is called the *Clausius–Mossotti equation*, since Mossotti (in 1850) and Clausius independently (in 1879) established that for any given substance $(\epsilon/\epsilon_0 - 1)/(\epsilon/\epsilon_0 + 2)$ should be proportional to the density of the substance.* The relation holds best for dilute substances such as gases. For liquids and solids, (4.70) is only approximately valid, especially if the dielectric constant is large. The interested reader can refer to the books by *Böttcher*, *Debye*, and *Fröhlich* for further details.

## *4.6 Models for the Molecular Polarizability*

The polarization of a collection of atoms or molecules can arise in two ways:

the applied field distorts the charge distributions and so produces an induced dipole moment in each molecule;

the applied field tends to line up the initially randomly oriented permanent dipole moments of the molecules.

To estimate the induced moments we consider a simple model of harmonically bound charges (electrons and ions). Each charge $e$ is bound under the action of a restoring force

$$\mathbf{F} = -m\omega_0^2\mathbf{x} \tag{4.71}$$

where $m$ is the mass of the charge, and $\omega_0$ the frequency of oscillation about equilibrium. Under the action of an electric field $\mathbf{E}$ the charge is displaced from its equilibrium by an amount $\mathbf{x}$ given by

$$m\omega_0^2\mathbf{x} = e\mathbf{E}$$

*At optical frequencies, $\epsilon/\epsilon_0 = n^2$, where $n$ is the index of refraction. With $n^2$ replacing $\epsilon/\epsilon_0$ in (4.70), the equation is sometimes called the *Lorenz–Lorentz equation* (1880).

Consequently the induced dipole moment is

$$\mathbf{p}_{\text{mol}} = e\mathbf{x} = \frac{e^2}{m\omega_0^2}\mathbf{E} \tag{4.72}$$

This means that the polarizability is $\gamma = e^2/m\omega_0^2\epsilon_0$. If there are a set of charges $e_j$ with masses $m_j$ and oscillation frequencies $\omega_j$ in each molecule then the molecular polarizability is

$$\gamma_{\text{mol}} = \frac{1}{\epsilon_0}\sum_j \frac{e_j^2}{m_j\omega_j^2} \tag{4.73}$$

To get a feeling for the order of magnitude of $\gamma$ we can make two different estimates. Since $\gamma$ has the dimensions of a volume, its magnitude must be of the order of molecular dimensions or less, namely $\gamma_{\text{el}} \lesssim 10^{-29}$ m$^3$. Alternatively, we note that the binding frequencies of electrons in atoms must be of the order of light frequencies. Taking a typical wavelength of light as 3000 Å, we find $\omega \simeq 6 \times 10^{15}$ s$^{-1}$. Then the electronic contribution to $\gamma$ is $\gamma_{\text{el}} \sim (e^2/m\omega^2\epsilon_0) \sim 0.88 \times 10^{-29}$ m$^3$, consistent with the molecular volume estimate. For gases at NTP the number of molecules per cubic meter is $N = 2.7 \times 10^{25}$, so that their susceptibilities should be of the order of $\chi_e \lesssim 10^{-3}$. This means dielectric constants differing from unity by a few parts in $10^3$, or less. Experimentally, typical values of dielectric constant are 1.00054 for air, 1.0072 for ammonia vapor, 1.0057 for methyl alcohol, 1.000068 for helium. For solid or liquid dielectrics, $N \sim 10^{28} - 10^{29}$ molecules/m$^3$. Consequently, the susceptibility can be of the order of unity (to within a factor $10^{\pm 1}$) as is observed.*

The possibility that thermal agitation of the molecules could modify the result (4.73) for the induced dipole polarizability needs consideration. In statistical mechanics the probability distribution of particles in phase space ($\mathbf{p}$, $\mathbf{q}$ space) is some function $f(H)$ of the Hamiltonian. For classical systems,

$$f(H) = e^{-H/kT} \tag{4.74}$$

is the Boltzmann factor. For the simple problem of the harmonically bound charge with an applied field in the $z$ direction, the Hamiltonian is

$$H = \frac{1}{2m}\mathbf{p}^2 + \frac{m}{2}\omega_0^2\mathbf{x}^2 - eEz \tag{4.75}$$

where here $\mathbf{p}$ is the momentum of the charged particle. The average value of the dipole moment in the $z$ direction is

$$\langle p_{\text{mol}}\rangle = \frac{\int d^3p \int d^3x\, (ez) f(H)}{\int d^3p \int d^3x\, f(H)} \tag{4.76}$$

If we introduce a displaced coordinate $\mathbf{x}' = \mathbf{x} - eE\hat{\mathbf{z}}/m\omega_0^2$ then

$$H = \frac{1}{2m}\mathbf{p}^2 + \frac{m\omega_0^2}{2}(\mathbf{x}')^2 - \frac{e^2E^2}{2m\omega_0^2} \tag{4.77}$$

*See, e.g., *CRC Handbook of Chemistry and Physics*, 78th ed., ed. D. R. Lipe, CRC Press, Boca Raton, FL (1997–98).

and

$$\langle p_{\text{mol}} \rangle = \frac{\int d^3p \int d^3x' \left( ez' + \dfrac{e^2 E}{m\omega_0^2} \right) f H)}{\int d^3p \int d^3x' \; f(H)} \tag{4.78}$$

Since $H$ is even in $z'$ the first integral vanishes. Thus, independent of the form of $f(H)$, we obtain

$$\langle p_{\text{mol}} \rangle = \frac{e^2}{m\omega_0^2} E$$

just as was found in (4.72), ignoring thermal motion.

The second type of polarizability is that caused by the partial orientation of otherwise random permanent dipole moments. This orientation polarization is important in "polar" substances such as HCl and $H_2O$ and was first discussed by Debye (1912). All molecules are assumed to possess a permanent dipole moment $\mathbf{p}_0$, which can be oriented in any direction in space. In the absence of a field, thermal agitation keeps the molecules randomly oriented so that there is no net dipole moment. With an applied field there is a tendency to line up along the field in the configuration of lowest energy. Consequently there will be an average dipole moment. To calculate this we note that the Hamiltonian of the molecule is given by

$$H = H_0 - \mathbf{p}_0 \cdot \mathbf{E} \tag{4.79}$$

where $H_0$ is a function of only the "internal" coordinates of the molecule. Using the Boltzmann factor (4.74), we can write the average dipole moment as:

$$\langle p_{\text{mol}} \rangle = \frac{\int d\Omega \; p_0 \cos\theta \exp\left( \dfrac{p_0 E \cos\theta}{kT} \right)}{\int d\Omega \; \exp\left( \dfrac{p_0 E \cos\theta}{kT} \right)} \tag{4.80}$$

where we have chosen $\mathbf{E}$ along the $z$ axis, integrated out all the irrelevant variables, and noted that only the component of $\langle \mathbf{p}_0 \rangle$ parallel to the field is different from zero. In general, $(p_0 E/kT)$ is very small compared to unity, except at low temperatures. Hence we can expand the exponentials and obtain the result:

$$\langle p_{\text{mol}} \rangle \simeq \frac{1}{3} \frac{p_0^2}{kT} E \tag{4.81}$$

The orientation polarization depends inversely on the temperature, as might be expected of an effect in which the applied field must overcome the opposition of thermal agitation.

In general both types of polarization, induced (electronic and ionic) and orientation, are present, and the general form of the molecular polarization is

$$\gamma_{\text{mol}} \simeq \gamma_i + \frac{1}{3\epsilon_0} \frac{p_0^2}{kT} \tag{4.82}$$

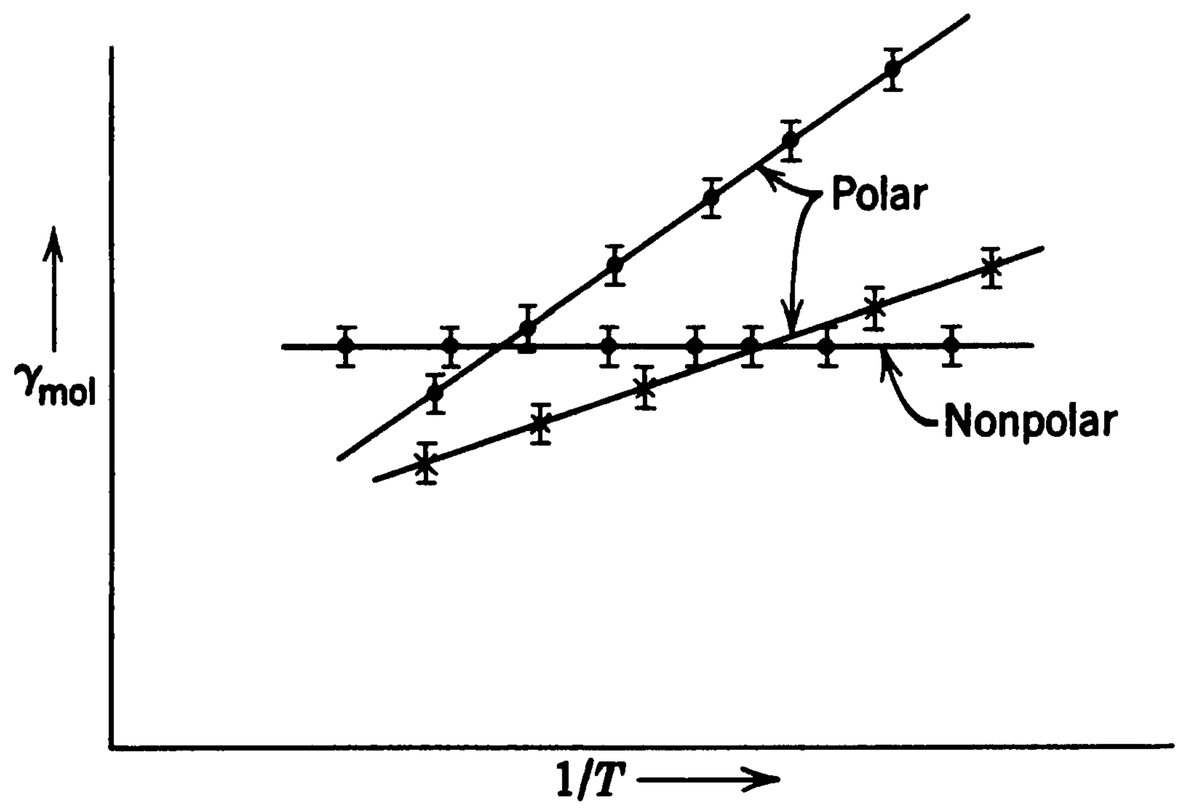



**Figure 4.10** Variation of molecular polarizability $\gamma_{mol}$ with temperature for polar and nonpolar substances: $\gamma_{mol}$ versus $T^{-1}$.

This shows a temperature dependence of the form $(a + b/T)$ so that the two types of polarization can be separated experimentally, as indicated in Fig. 4.10. For "polar" molecules, such as HCl and $H_2O$, the observed permanent dipole moments are of the order of an electronic charge times $10^{-8}$ cm, in accordance with molecular dimensions.

## 4.7 *Electrostatic Energy in Dielectric Media*

In Section 1.11 we discussed the energy of a system of charges in free space. The result obtained there,

$$W = \frac{1}{2} \int \rho(\mathbf{x})\Phi(\mathbf{x})\, d^3x \tag{4.83}$$

for the energy due to a charge density $\rho(\mathbf{x})$ and a potential $\Phi(\mathbf{x})$ cannot in general be taken over as it stands in our macroscopic description of dielectric media. The reason becomes clear when we recall how (4.83) was obtained. We thought of the final configuration of charge as being created by assembling bit by bit the elemental charges, bringing each one in from infinitely far away against the action of the then existing electric field. The total work done was given by (4.83). With dielectric media, work is done not only to bring real (macroscopic) charge into position, but also to produce a certain state of polarization in the medium. If $\rho$ and $\Phi$ in (4.83) represent macroscopic variables, it is certainly not evident that (4.83) represents the total work, including that done on the dielectric.

To be general in our description of dielectrics, we will not initially make any assumptions about linearity, uniformity, etc., of the response of a dielectric to an applied field. Rather, let us consider a small change in the energy $\delta W$ due to some sort of change $\delta\rho$ in the macroscopic charge density $\rho$ existing in all space. The work done to accomplish this change is

$$\delta W = \int \delta\rho(\mathbf{x})\Phi(\mathbf{x})\, d^3x \tag{4.84}$$

where $\Phi(\mathbf{x})$ is the potential due to the charge density $\rho(\mathbf{x})$ already present. Since $\nabla \cdot \mathbf{D} = \rho$, we can relate the change $\delta\rho$ to a change in the displacement of $\delta\mathbf{D}$:

$$\delta\rho = \nabla \cdot (\delta\mathbf{D}) \tag{4.85}$$

Then the energy change $\delta W$ can be cast into the form:

$$\delta W = \int \mathbf{E} \cdot \delta\mathbf{D}\, d^3x \tag{4.86}$$

where we have used $\mathbf{E} = -\nabla\Phi$ and have assumed that $\rho(\mathbf{x})$ was a localized charge distribution. The total electrostatic energy can now be written down formally, at least, by allowing $\mathbf{D}$ to be brought from an initial value $\mathbf{D} = 0$ to its final value $\mathbf{D}$:

$$W = \int d^3x \int_0^D \mathbf{E} \cdot \delta\mathbf{D} \tag{4.87}$$

If the medium is *linear*, then

$$\mathbf{E} \cdot \delta\mathbf{D} = \tfrac{1}{2}\delta(\mathbf{E} \cdot \mathbf{D}) \tag{4.88}$$

and the total electrostatic energy is

$$W = \frac{1}{2}\int \mathbf{E} \cdot \mathbf{D}\, d^3x \tag{4.89}$$

This last result can be transformed into (4.83) by using $\mathbf{E} = -\nabla\Phi$ and $\nabla \cdot \mathbf{D} = \rho$, or by going back to (4.84) and assuming that $\rho$ and $\Phi$ are connected linearly. Thus we see that (4.83) *is valid macroscopically only if the behavior is linear.* Otherwise the energy of a final configuration must be calculated from (4.87) and might conceivably depend on the past history of the system (hysteresis effects).

A problem of considerable interest is the change in energy when a dielectric object with a linear response is placed in an electric field whose sources are fixed. Suppose that initially the electric field $\mathbf{E}_0$ due to a certain distribution of charges $\rho_0(\mathbf{x})$ exists in a medium of electric susceptibility $\epsilon_0$, which may be a function of position (for the moment $\epsilon_0$ is not the susceptibility of the vacuum). The initial electrostatic energy is

$$W_0 = \frac{1}{2}\int \mathbf{E}_0 \cdot \mathbf{D}_0\, d^3x$$

where $\mathbf{D}_0 = \epsilon_0\mathbf{E}_0$. Then with the sources fixed in position a dielectric object of volume $V_1$ is introduced into the field, changing the field from $\mathbf{E}_0$ to $\mathbf{E}$. The presence of the object can be described by a susceptibility $\epsilon(\mathbf{x})$, which has the value $\epsilon_1$ inside $V_1$ and $\epsilon_0$ outside $V_1$. To avoid mathematical difficulties we can imagine $\epsilon(\mathbf{x})$ to be a smoothly varying function of position that falls rapidly but continuously from $\epsilon_1$ to $\epsilon_0$ at the edge of the volume $V_1$. The energy now has the value

$$W_1 = \frac{1}{2}\int \mathbf{E} \cdot \mathbf{D}\, d^3x$$

where $\mathbf{D} = \epsilon\mathbf{E}$. The difference in the energy can be written:

$$\begin{aligned} W &= \frac{1}{2}\int (\mathbf{E} \cdot \mathbf{D} - \mathbf{E}_0 \cdot \mathbf{D}_0)\, d^3x \\ &= \frac{1}{2}\int (\mathbf{E} \cdot \mathbf{D}_0 - \mathbf{D} \cdot \mathbf{E}_0)\, d^3x + \frac{1}{2}\int (\mathbf{E} + \mathbf{E}_0) \cdot (\mathbf{D} - \mathbf{D}_0)\, d^3x \end{aligned} \tag{4.90}$$

The second integral can be shown to vanish by the following argument. Since $\nabla \times (\mathbf{E} + \mathbf{E}_0) = 0$, we can write

$$\mathbf{E} + \mathbf{E}_0 = -\nabla\Phi$$

Then the second integral becomes:

$$I = -\frac{1}{2}\int \nabla\Phi \cdot (\mathbf{D} - \mathbf{D}_0)\, d^3x$$

Integration by parts transforms this into

$$I = \frac{1}{2}\int \Phi\nabla \cdot (\mathbf{D} - \mathbf{D}_0)\, d^3x = 0$$

since $\nabla \cdot (\mathbf{D} - \mathbf{D}_0) = 0$ because the source charge density $\rho_0(\mathbf{x})$ is assumed unaltered by the insertion of the dielectric object. Consequently the energy change is

$$W = \frac{1}{2}\int (\mathbf{E} \cdot \mathbf{D}_0 - \mathbf{D} \cdot \mathbf{E}_0)\, d^3x \tag{4.91}$$

The integration appears to be over all space, but is actually only over the volume $V_1$ of the object, since, outside $V_1$, $\mathbf{D} = \epsilon_0\mathbf{E}$. Therefore we can write

$$W = -\frac{1}{2}\int_{V_1} (\epsilon_1 - \epsilon_0)\mathbf{E} \cdot \mathbf{E}_0\, d^3x \tag{4.92}$$

If the medium surrounding the dielectric body is free space, then using the definition of polarization $\mathbf{P}$, (4.92) can then be expressed in the form:

$$W = -\frac{1}{2}\int_{V_1} \mathbf{P} \cdot \mathbf{E}_0\, d^3x \tag{4.93}$$

where $\mathbf{P}$ is the polarization of the dielectric. This shows that the energy density of a dielectric placed in a field $\mathbf{E}_0$ whose sources are fixed is given by

$$w = -\tfrac{1}{2}\mathbf{P} \cdot \mathbf{E}_0 \tag{4.94}$$

This result is analogous to the dipole term in the energy (4.24) of a charge distribution in an external field. The factor $\frac{1}{2}$ is due to the fact that (4.94) represents the energy density of a polarizable dielectric in an external field, rather than a permanent dipole. It is the same factor $\frac{1}{2}$ that appears in (4.88).

Equations (4.92) and (4.93) show that a dielectric body will tend to move toward regions of increasing field $\mathbf{E}_0$ provided $\epsilon_1 > \epsilon_0$. To calculate the force acting we can imagine a small generalized displacement of the body $\delta\xi$. Then there will be a change in the energy $\delta W$. Since the charges are held fixed, there is no external source of energy and the change in field energy can be interpreted as a change in the potential energy of the body. This means that there is a force acting on the body:

$$F_\xi = -\left(\frac{\partial W}{\partial \xi}\right)_Q \tag{4.95}$$

where the subscript $Q$ has been placed on the partial derivative to indicate that *the sources of the field are kept fixed.*

In practical situations involving the motion of dielectrics the electric fields are often produced by a configuration of electrodes held at *fixed potentials* by connection to an external source such as a battery. To maintain the potentials constant as the distribution of dielectric varies, charge will flow to or from the battery to the electrodes. This means that energy is being supplied from the external source, and it is of interest to compare the energy supplied in that way with the energy change found above *for fixed sources* of the field. We will treat only linear media so that (4.83) is valid. It is sufficient to consider small changes in an existing configuration. From (4.83) it is evident that the change in energy accompanying the changes $\delta\rho(\mathbf{x})$ and $\delta\Phi(\mathbf{x})$ in charge density and potential is

$$\delta W = \frac{1}{2}\int (\rho\ \delta\Phi + \Phi\ \delta\rho)\ d^3x \tag{4.96}$$

Comparison with (4.84) shows that, if the dielectric properties are not changed, the two terms in (4.96) are equal. If, however, the dielectric properties are altered,

$$\epsilon(\mathbf{x}) \to \epsilon(\mathbf{x}) + \delta\epsilon(\mathbf{x}) \tag{4.97}$$

the contributions in (4.96) are not necessarily the same. In fact, we have just calculated the change in energy brought about by introducing a dielectric body into an electric field whose sources were fixed ($\delta\rho = 0$). Equal contributions in (4.96) would imply $\delta W = 0$, but (4.91) or (4.92) are not zero in general. The reason for this difference lies in the existence of the polarization charge. The change in dielectric properties implied by (4.97) can be thought of as a change in the polarization-charge density. If then (4.96) is interpreted as an integral over both free and polarization-charge densities (i.e., a microscopic equation), the two contributions are always equal. However, it is often convenient to deal with macroscopic quantities. Then the equality holds only if the dielectric properties are unchanged.

The process of altering the dielectric properties in some way (by moving the dielectric bodies, by changing their susceptibilities, etc.) in the presence of electrodes at fixed potentials can be viewed as taking place in two steps. In the first step the electrodes are disconnected from the batteries and the charges on them held fixed ($\delta\rho = 0$). With the change (4.97) in dielectric properties, the energy change is

$$\delta W_1 = \frac{1}{2}\int \rho\ \delta\Phi_1\ d^3x \tag{4.98}$$

where $\delta\Phi_1$ is the change in potential produced. This can be shown to yield the result (4.92). In the second step the batteries are connected again to the electrodes to restore their potentials to the original values. There will be a flow of charge $\delta\rho_2$ from the batteries accompanying the change in potential* $\delta\Phi_2 = -\delta\Phi_1$. Therefore the energy change in the second step is

$$\delta W_2 = \frac{1}{2}\int (\rho\ \delta\Phi_2 + \Phi\ \delta\rho_2)\ d^3x = -2\delta W_1 \tag{4.99}$$

*Note that it is necessary merely to know that $\delta\Phi_2 = -\delta\Phi_1$ on the electrodes, since that is the only place where free charge resides.

since the two contributions are equal. In the second step we find the external sources changing the energy in the opposite sense and by twice the amount of the initial step. Consequently the net change is

$$\delta W = -\frac{1}{2}\int \rho\, \delta\Phi_1\, d^3x \tag{4.100}$$

Symbolically

$$\delta W_V = -\delta W_Q \tag{4.101}$$

where the subscript denotes the quantity held fixed. If a dielectric with $\epsilon/\epsilon_0 > 1$ moves into a region of greater field strength, the energy increases instead of decreases. For a generalized displacement $d\xi$ the mechanical force acting is now

$$F_\xi = +\left(\frac{\partial W}{\partial \xi}\right)_V \tag{4.102}$$

## *References and Suggested Reading*

The derivation of the macroscopic equations of electrostatics by averaging over aggregates of atoms is presented in Chapter 6 and by

Rosenfeld, Chapter II
Mason and Weaver, Chapter I, Part III
Van Vleck, Chapter 1

Rosenfeld also treats the classical electron theory of dielectrics. Van Vleck's book is devoted to electric and magnetic susceptibilities. Specific works on electric polarization phenomena are those of

Böttcher
Debye
Fröhlich

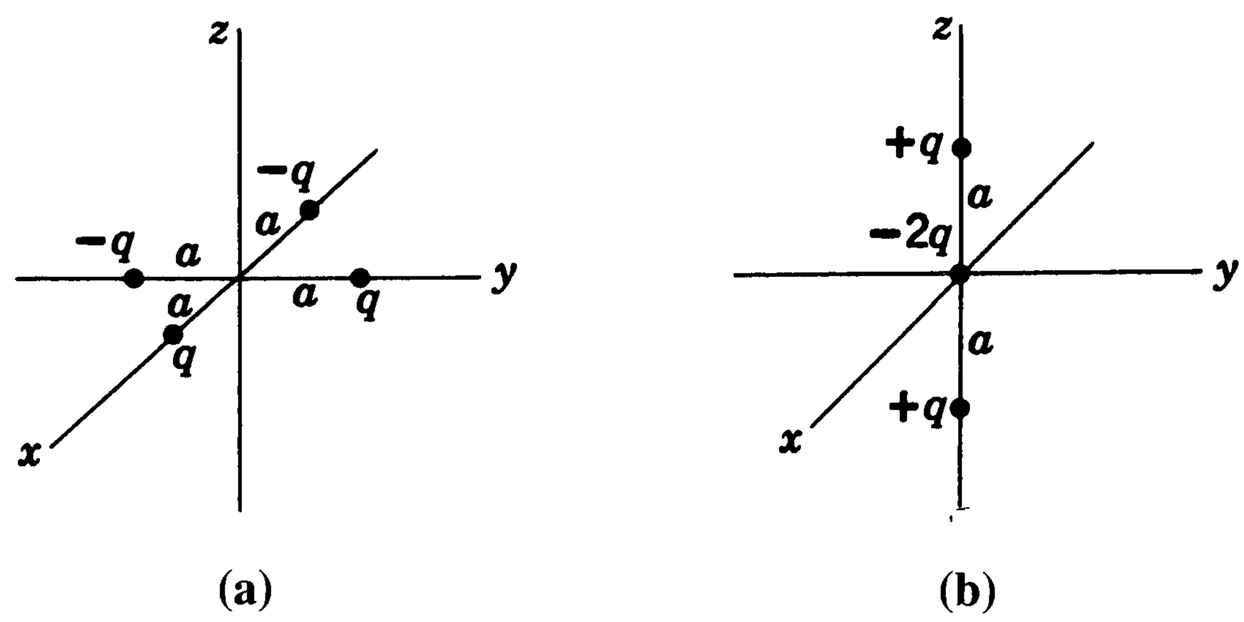
Boundary-value problems with dielectrics are discussed in all the references on electrostatics in Chapters 2 and 3.

Our treatment of forces and energy with dielectric media is brief. More extensive discussions, including forces on liquid and solid dielectrics, the electric stress tensor, electrostriction, and thermodynamic effects, may be found in

Abraham and Becker, Band 1, Chapter V
Durand, Chapters VI and VII
Landau and Lifshitz, *Electrodynamics of Continuous Media*
Maxwell, Vol. 1, Chapter V
Panofsky and Phillips, Chapter 6
Stratton, Chapter II

## *Problems*

**4.1** Calculate the multipole moments $q_{lm}$ of the charge distributions shown as parts a and b. Try to obtain results for the nonvanishing moments valid for all $l$, but in each case find the first *two* sets of nonvanishing moments at the very least.




**Problem 4.1**

**(c)** For the charge distribution of the second set b write down the multipole expansion for the potential. Keeping only the lowest-order term in the expansion, plot the potential in the $x$-$y$ plane as a function of distance from the origin for distances greater than $a$.

**(d)** Calculate directly from Coulomb's law the exact potential for b in the $x$-$y$ plane. Plot it as a function of distance and compare with the result found in part c.

Divide out the asymptotic form in parts c and d to see the behavior at large distances more clearly.

**4.2** A point dipole with dipole moment $\mathbf{p}$ is located at the point $\mathbf{x}_0$. From the properties of the derivative of a Dirac delta function, show that for calculation of the potential $\Phi$ or the energy of a dipole in an external field, the dipole can be described by an effective charge density

$$\rho_{\text{eff}}(\mathbf{x}) = -\mathbf{p} \cdot \nabla\delta(\mathbf{x} - \mathbf{x}_0)$$

**4.3** The $l$th term in the multipole expansion (4.1) of the potential is specified by the $(2l + 1)$ multipole moments $q_{lm}$. On the other hand, the Cartesian multipole moments,

$$Q^{(l)}_{\alpha\beta\gamma} = \int \rho(\mathbf{x}) x^\alpha y^\beta z^\gamma \, d^3x$$

with $\alpha$, $\beta$, $\gamma$ nonnegative integers subject to the constraint $\alpha + \beta + \gamma = l$, are $(l + 1)(l + 2)/2$ in number. Thus, for $l > 1$ there are more Cartesian multipole moments than seem necessary to describe the term in the potential whose radial dependence is $r^{-l-1}$.

Show that while the $q_{lm}$ transform under rotations as irreducible spherical tensors of rank $l$, the Cartesian multipole moments correspond to reducible spherical tensors of ranks $l, l - 2, l - 4, \ldots, l_{\min}$, where $l_{\min} = 0$ or 1 for $l$ even or odd, respectively. Check that the number of different tensorial components adds up to the total number of Cartesian tensors. Why are only the $q_{lm}$ needed in the expansion (4.1)?

**4.4** **(a)** Prove the following theorem: For an arbitrary charge distribution $\rho(\mathbf{x})$ the values of the $(2l + 1)$ moments of the first nonvanishing multipole are independent of the origin of the coordinate axes, but the values of all higher multipole moments do in general depend on the choice of origin. (The different moments $q_{lm}$ for fixed $l$ depend, of course, on the orientation of the axes.)

**(b)** A charge distribution has multipole moments $q$, $\mathbf{p}$, $Q_{ij}, \ldots$ with respect to one set of coordinate axes, and moments $q'$, $\mathbf{p}'$, $Q'_{ij}, \ldots$ with respect to another

set whose axes are parallel to the first, but whose origin is located at the point $\mathbf{R} = (X, Y, Z)$ relative to the first. Determine explicitly the connections between the monopole, dipole, and quadrupole moments in the two coordinate frames.

**(c)** If $q \neq 0$, can $\mathbf{R}$ be found so that $\mathbf{p}' = 0$? If $q \neq 0$, $\mathbf{p} \neq 0$, or at least $\mathbf{p} \neq 0$, can $\mathbf{R}$ be found so that $Q'_{ij} = 0$?

**4.5** A localized charge density $\rho(x, y, z)$ is placed in an external electrostatic field described by a potential $\Phi^{(0)}(x, y, z)$. The external potential varies slowly in space over the region where the charge density is different from zero.

**(a)** From first principles calculate the total *force* acting on the charge distribution as an expansion in multipole moments times derivatives of the electric field, up to and including the quadrupole moments. Show that the force is

$$\mathbf{F} = q\mathbf{E}^{(0)}(0) + \{\nabla[\mathbf{p} \cdot \mathbf{E}^{(0)}(\mathbf{x})]\}_0 + \left\{\nabla\left[\frac{1}{6}\sum_{j,k} Q_{jk}\frac{\partial E_j^{(0)}}{\partial x_k}(\mathbf{x})\right]\right\}_0 + \cdots$$

Compare this to the expansion (4.24) of the *energy* $W$. Note that (4.24) is a number—it is not a function of $\mathbf{x}$ that can be differentiated! What is its connection to $\mathbf{F}$?

**(b)** Repeat the calculation of part a for the total *torque*. For simplicity, evaluate only one Cartesian component of the torque, say $N_1$. Show that this component is

$$N_1 = [\mathbf{p} \times \mathbf{E}^{(0)}(0)]_1 + \frac{1}{3}\left[\frac{\partial}{\partial x_3}\left(\sum_j Q_{2j}E_j^{(0)}\right) - \frac{\partial}{\partial x_2}\left(\sum_j Q_{3j}E_j^{(0)}\right)\right]_0 + \cdots$$

**4.6** A nucleus with quadrupole moment $Q$ finds itself in a cylindrically symmetric electric field with a gradient $(\partial E_z/\partial z)_0$ along the $z$ axis at the position of the nucleus.

**(a)** Show that the energy of quadrupole interaction is

$$W = -\frac{e}{4}Q\left(\frac{\partial E_z}{\partial z}\right)_0$$

**(b)** If it is known that $Q = 2 \times 10^{-28}$ m$^2$ and that $W/h$ is 10 MHz, where $h$ is Planck's constant, calculate $(\partial E_z/\partial z)_0$ in units of $e/4\pi\epsilon_0 a_0^3$, where $a_0 = 4\pi\epsilon_0\hbar^2/me^2 = 0.529 \times 10^{-10}$ m is the Bohr radius in hydrogen.

**(c)** Nuclear charge distributions can be approximated by a constant charge density throughout a spheroidal volume of semimajor axis $a$ and semiminor axis $b$. Calculate the quadrupole moment of such a nucleus, assuming that the total charge is $Ze$. Given that $Eu^{153}$ ($Z = 63$) has a quadrupole moment $Q = 2.5 \times 10^{-28}$ m$^2$ and a mean radius

$$R = (a + b)/2 = 7 \times 10^{-15} \text{ m}$$

determine the fractional difference in radius $(a - b)/R$.

**4.7** A localized distribution of charge has a charge density

$$\rho(\mathbf{r}) = \frac{1}{64\pi} r^2 e^{-r} \sin^2\theta$$

**(a)** Make a multipole expansion of the potential due to this charge density and determine all the nonvanishing multipole moments. Write down the potential at large distances as a finite expansion in Legendre polynomials.

**(b)** Determine the potential explicitly at any point in space, and show that near the origin, correct to $r^2$ inclusive,

$$\Phi(\mathbf{r}) \simeq \frac{1}{4\pi\epsilon_0}\left[\frac{1}{4} - \frac{r^2}{120}P_2(\cos\theta)\right]$$

**(c)** If there exists at the origin a nucleus with a quadrupole moment $Q = 10^{-28}\,\mathrm{m}^2$, determine the magnitude of the interaction energy, assuming that the unit of charge in $\rho(\mathbf{r})$ above is the electronic charge and the unit of length is the hydrogen Bohr radius $a_0 = 4\pi\epsilon_0\hbar^2/me^2 = 0.529 \times 10^{-10}$ m. Express your answer as a frequency by dividing by Planck's constant $h$.

The charge density in this problem is that for the $m = \pm 1$ states of the $2p$ level in hydrogen, while the quadrupole interaction is of the same order as found in molecules.

**4.8** A very long, right circular, cylindrical shell of dielectric constant $\epsilon/\epsilon_0$ and inner and outer radii $a$ and $b$, respectively, is placed in a previously uniform electric field $E_0$ with its axis perpendicular to the field. The medium inside and outside the cylinder has a dielectric constant of unity.

**(a)** Determine the potential and electric field in the three regions, neglecting end effects.

**(b)** Sketch the lines of force for a typical case of $b \simeq 2a$.

**(c)** Discuss the limiting forms of your solution appropriate for a solid dielectric cylinder in a uniform field, and a cylindrical cavity in a uniform dielectric.

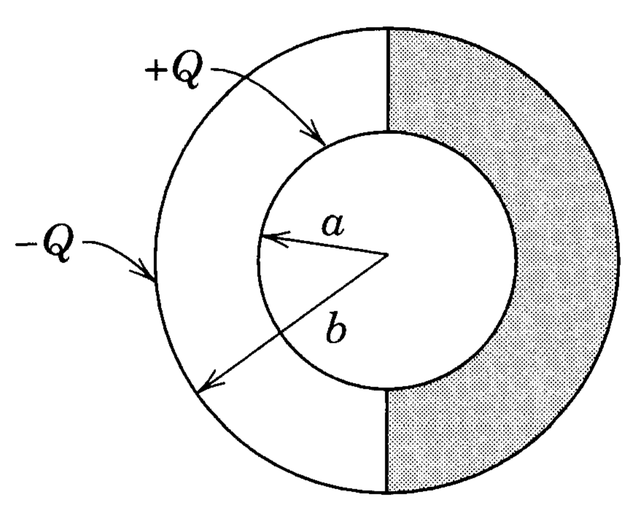
**4.9** A point charge $q$ is located in free space a distance $d$ from the center of a dielectric sphere of radius $a$ ($a < d$) and dielectric constant $\epsilon/\epsilon_0$.

**(a)** Find the potential at all points in space as an expansion in spherical harmonics.

**(b)** Calculate the rectangular components of the electric field *near* the center of the sphere.

**(c)** Verify that, in the limit $\epsilon/\epsilon_0 \to \infty$, your result is the same as that for the conducting sphere.

**4.10** Two concentric conducting spheres of inner and outer radii $a$ and $b$, respectively, carry charges $\pm Q$. The empty space between the spheres is half-filled by a hemispherical shell of dielectric (of dielectric constant $\epsilon/\epsilon_0$), as shown in the figure.




**Problem 4.10**

**(a)** Find the electric field everywhere between the spheres.

**(b)** Calculate the surface-charge distribution on the inner sphere.

**(c)** Calculate the polarization-charge density induced on the surface of the dielectric at $r = a$.

**4.11** The following data on the variation of dielectric constant with pressure are taken from the *Smithsonian Physical Tables*, 9th ed., p. 424:

| Air at 292 K | | |
|---|---|---|
| **Pressure (atm)** | $\epsilon/\epsilon_0$ | |
| 20 | 1.0108 | Relative density of air as a function of pressure is given in *AIP Handbook*, [3rd ed., McGraw-Hill, New York (1972), p. 4-165]. |
| 40 | 1.0218 | |
| 60 | 1.0333 | |
| 80 | 1.0439 | |
| 100 | 1.0548 | |

| Pentane ($C_5H_{12}$) at 303 K | | |
|---|---|---|
| **Pressure (atm)** | **Density (g/cm$^3$)** | $\epsilon/\epsilon_0$ |
| 1 | 0.613 | 1.82 |
| $10^3$ | 0.701 | 1.96 |
| $4 \times 10^3$ | 0.796 | 2.12 |
| $8 \times 10^3$ | 0.865 | 2.24 |
| $12 \times 10^3$ | 0.907 | 2.33 |

Test the Clausius–Mossotti relation between dielectric constants and density for air and pentane in the ranges tabulated. Does it hold exactly? Approximately? If approximately, discuss fractional variations in density and $(\epsilon/\epsilon_0 - 1)$. For pentane, compare the Clausius–Mossotti relation to the cruder relation, $(\epsilon/\epsilon_0 - 1) \propto$ density.

**4.12** Water vapor is a polar gas whose dielectric constant exhibits an appreciable temperature dependence. The following table gives experimental data on this effect. Assuming that water vapor obeys the ideal gas law, calculate the molecular polarizability as a function of inverse temperature and plot it. From the slope of the curve, deduce a value for the permanent dipole moment of the $H_2O$ molecule (express the dipole moment in coulomb-meters).

| T(K) | Pressure (cm Hg) | $(\epsilon/\epsilon_0 - 1) \times 10^5$ |
|---|---|---|
| 393 | 56.49 | 400.2 |
| 423 | 60.93 | 371.7 |
| 453 | 65.34 | 348.8 |
| 483 | 69.75 | 328.7 |

**4.13** Two long, coaxial, cylindrical conducting surfaces of radii $a$ and $b$ are lowered vertically into a liquid dielectric. If the liquid rises an average height $h$ between the electrodes when a potential difference $V$ is established between them, show that the susceptibility of the liquid is

$$\chi_e = \frac{(b^2 - a^2)\rho g h \ln(b/a)}{\epsilon_0 V^2}$$

where $\rho$ is the density of the liquid, $g$ is the acceleration due to gravity, and the susceptibility of air is neglected.